LAB

Build an AI agent using IBM watsonx.ai

IBM **SkillsBuild**

Contents

Introduction

In this lab, you'll design and deploy a personalized AI agent using IBM watsonx Agent Lab to support fitness, meal planning, and daily health guidance. This drag-and-drop tool enables you to build and validate a real-world healthcare solution without any coding experience.

Software requirements

To complete this lab, you'll need access to an IBM Cloud account with IBM watsonx.ai and Agent Lab enabled. These platforms provide the tools to build and deploy your AI agent. You'll also need a commonly used up to date web browser such as Chrome, Firefox, or Edge to access the lab environment smoothly.

Basic familiarity with AI concepts or low-code platforms might help you better understand the process.

Objective

After completing this lab, you should be able to:

- Build an AI agent using IBM watsonx.ai

Lab steps

This lab requires you to complete the following procedures:

- Log in to watsonx.ai
- Configure the project for Agent Lab
- Build and configure the Personal Health Advisor agent
- Deploy and validate the agent using visual tools

Estimated duration to complete

30 minutes

Scenario

Background information

A wellness startup called VitaWell provides digital health solutions to its users, including fitness tracking, workout routines, and meal planning. As its customer base grows, the company is exploring the use of a virtual AI agent to deliver personalized, always-available support tailored to each user's goals and health conditions. You've been hired as the lead AI designer to develop this AI-enabled Personal Health Advisor using watsonx Agent Lab.

Challenge

Currently, the team relies on manual content delivery of health advice through blogs and chat scripts. Health coaches are overworked, which leads to delayed response times, generic guidance, and limited user interaction. The company needs a more interactive, responsive, and personalized experience that adapts to each user's unique lifestyle and goals.

Solution

You'll use watsonx Agent Lab to build a dynamic AI agent powered by the llama-3-3-70B-instruct model. The agent will gather user information and preferences, retrieve contextual data using tools, generate health plans, provide meal suggestions, and curate fitness routines while maintaining a conversational tone and protecting users' private data.

Log in to watsonx.ai

Overview

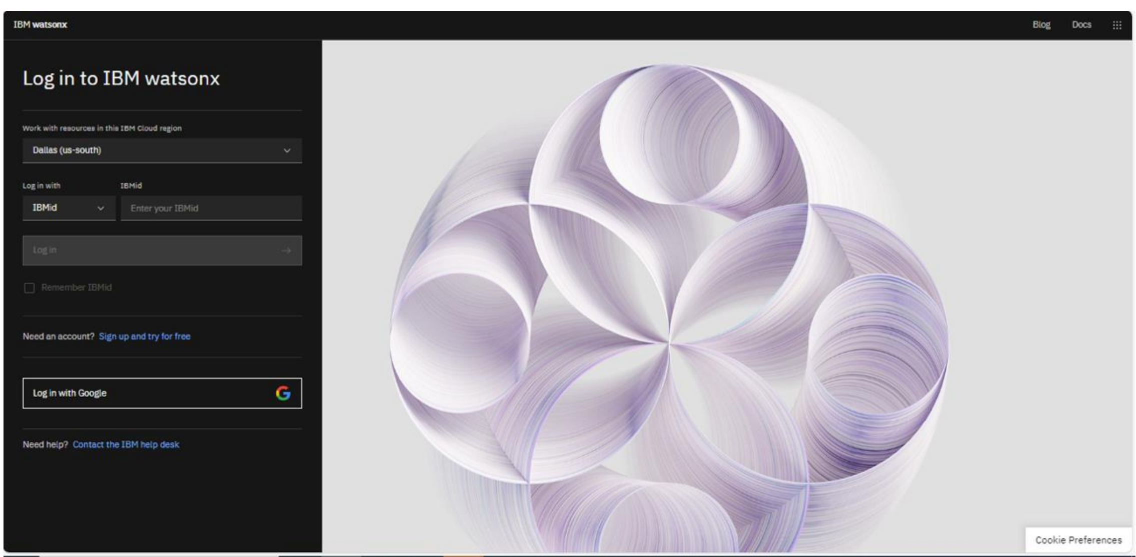
In this procedure, you'll create or sign in to your IBM Cloud account to access the watsonx.ai environment. IBM Cloud provides the infrastructure and tools needed to build, deploy, and manage AI solutions. You'll develop your AI agent on watsonx.ai, using built-in tools and features that enhance its ability to interpret prompts and respond to people. This setup ensures you have the environment and access needed to build and run the agent powered by the llama-3-3-70B-instruct model.

Instructions

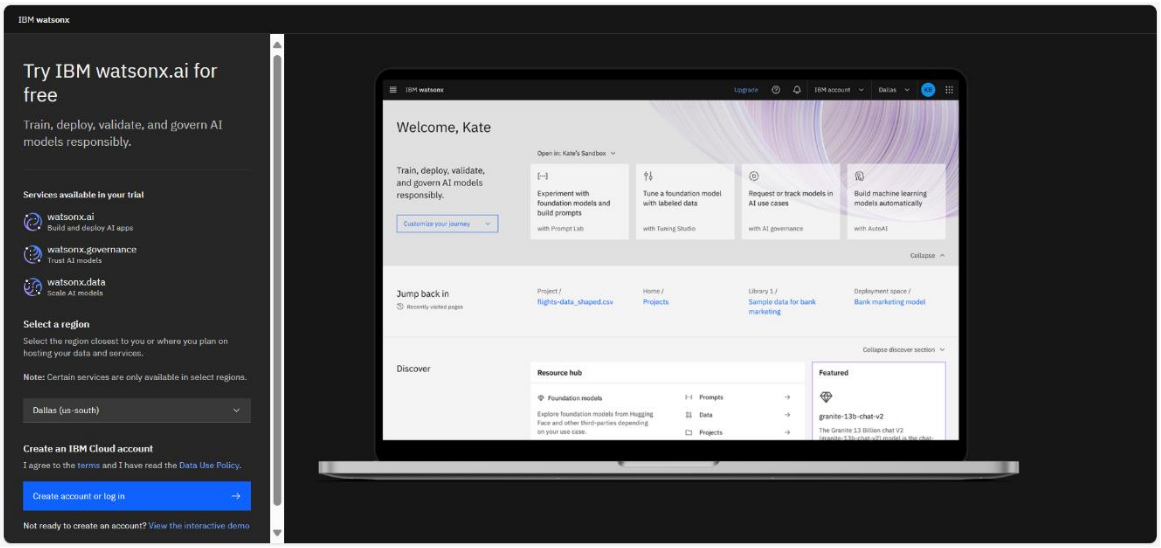
1. Go to the watsonx.ai platform.

2. Select Dallas(us-south) region.

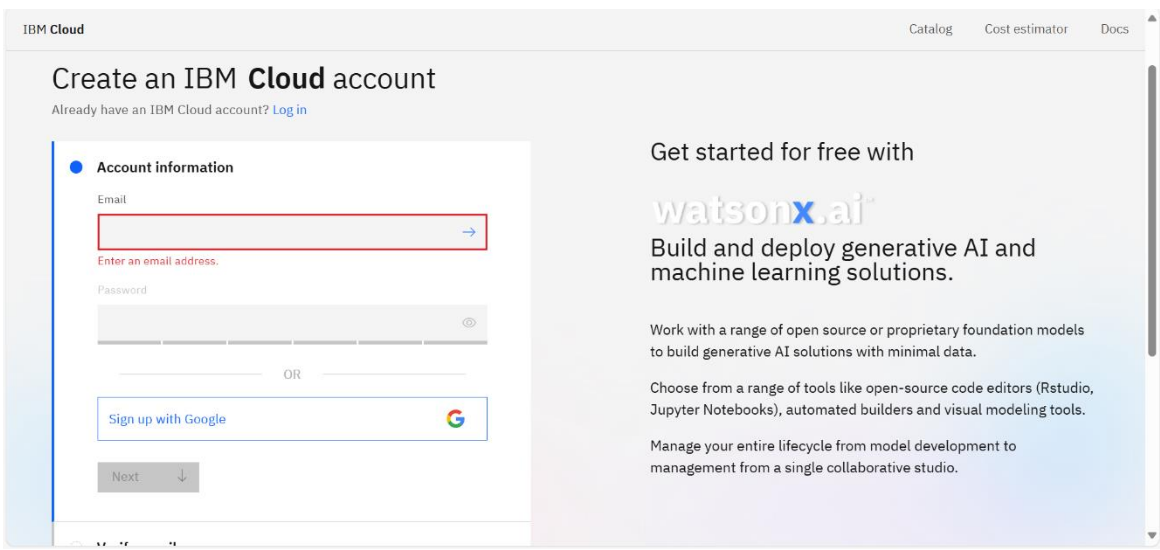
3. Select the **Sign up and try for free** link, as shown in the following screenshot.



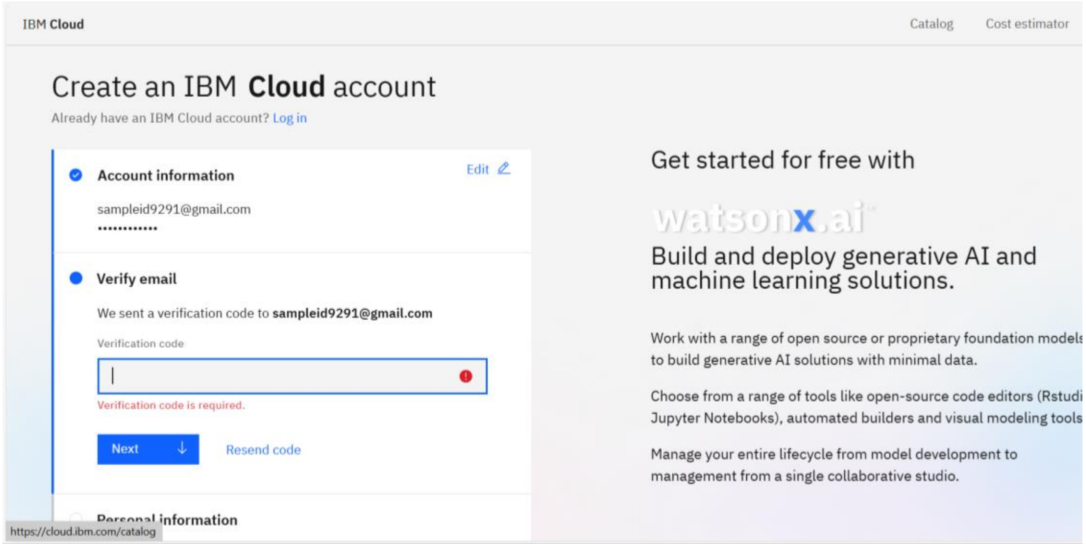
4. Select region as Dallas (us-south). Then, select **Create account or Log in**.



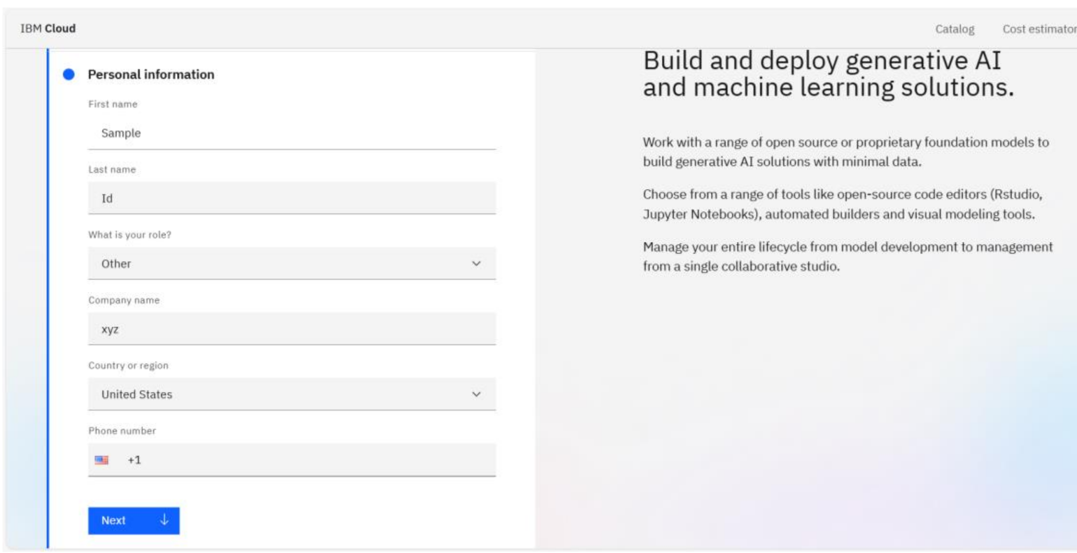
5. The IBM cloud platform opens as shown in the following screenshot. If you already have an IBM cloud account, select **Log in**. Else create an IBM cloud account using your email and password or select **Sign up with Google**.



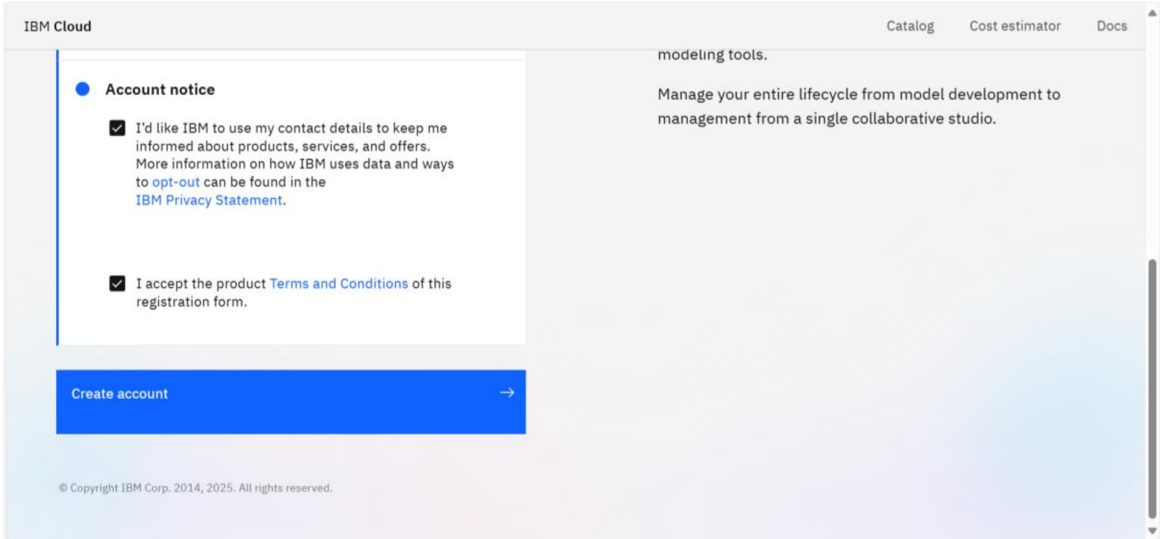
6. You'll receive a verification code on your email ID. Enter the code in the **Verification code** field and select **Next**.



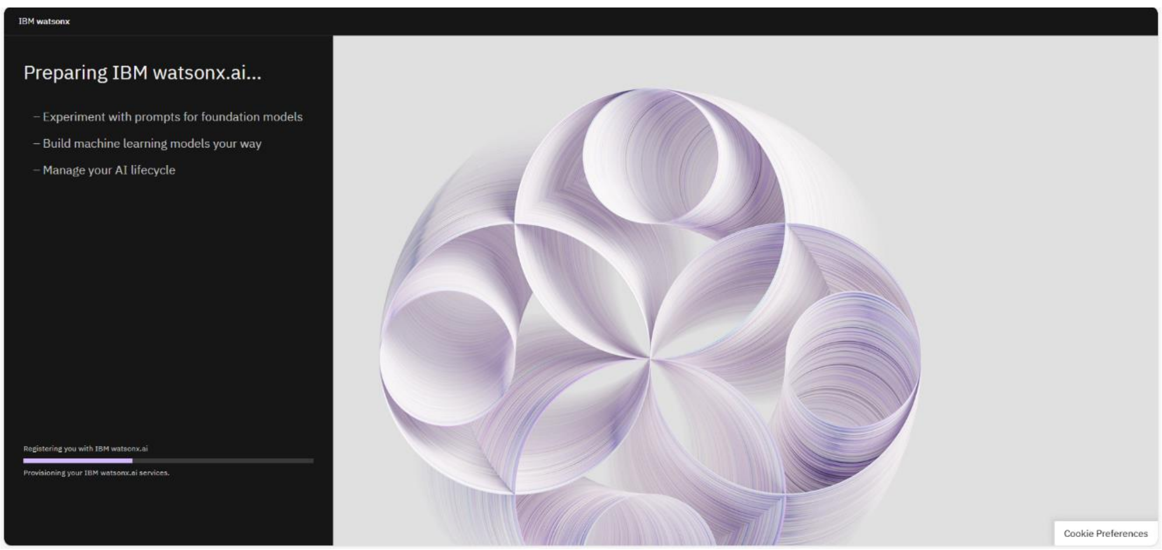
7. Enter your personal information as shown in the following screenshot. Then, select **Next**.



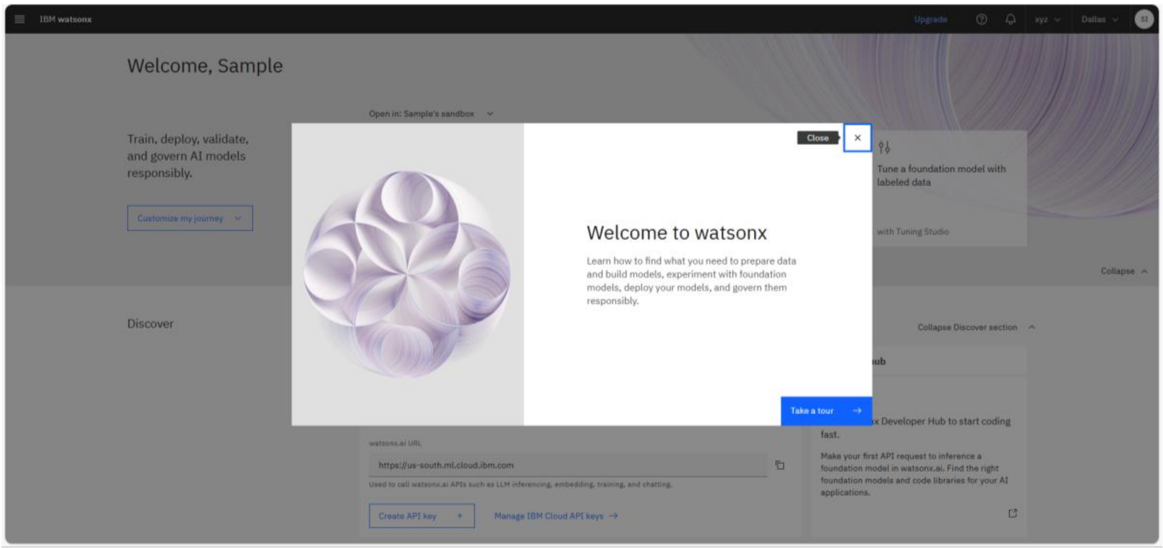
8. Accept the terms and conditions by selecting the recommended checkboxes and select **Create account** as shown in the following screenshot.



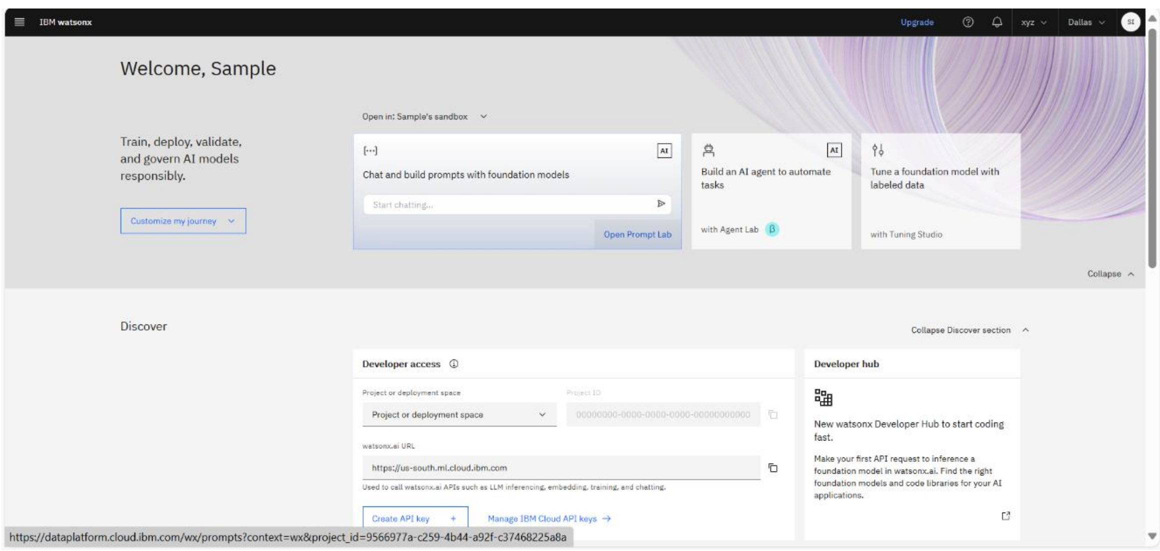
9. The IBM watsonx window opens, indicating that registration is in progress. Wait for a minute for it to be completed.



10. Welcome to watsonx dialog appears, select the **Close** button.



11. After you've signed in, the "watsonx.ai" page is displayed.



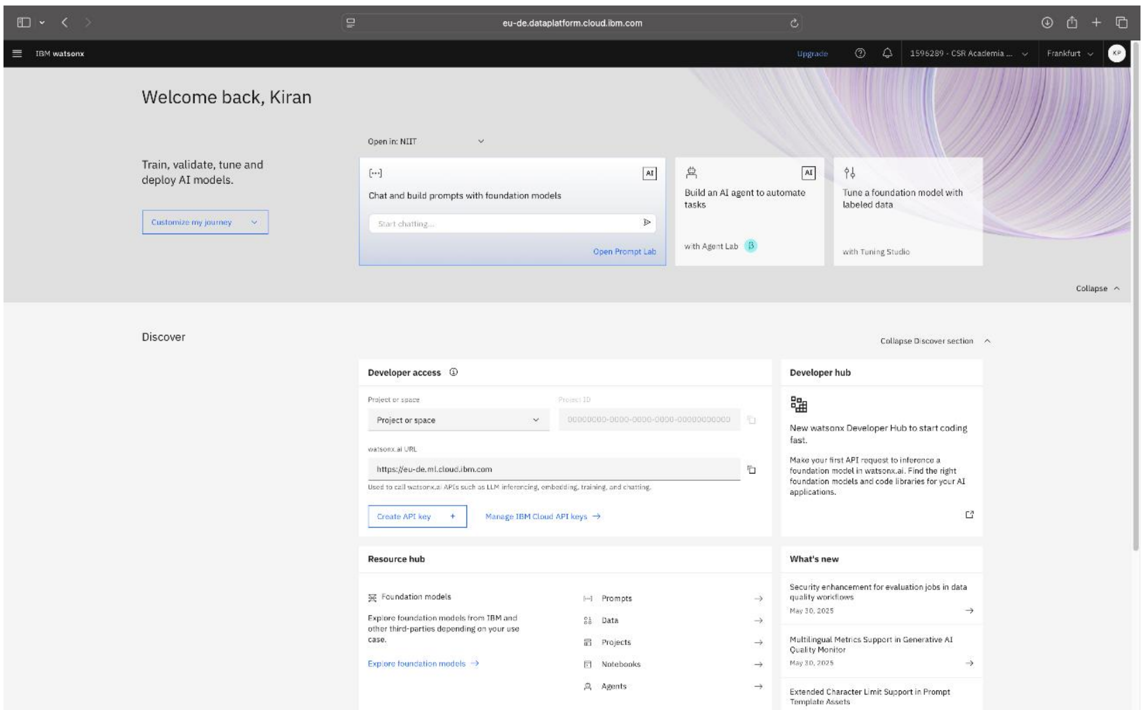
Configure the project for the Agent Lab environment

Overview

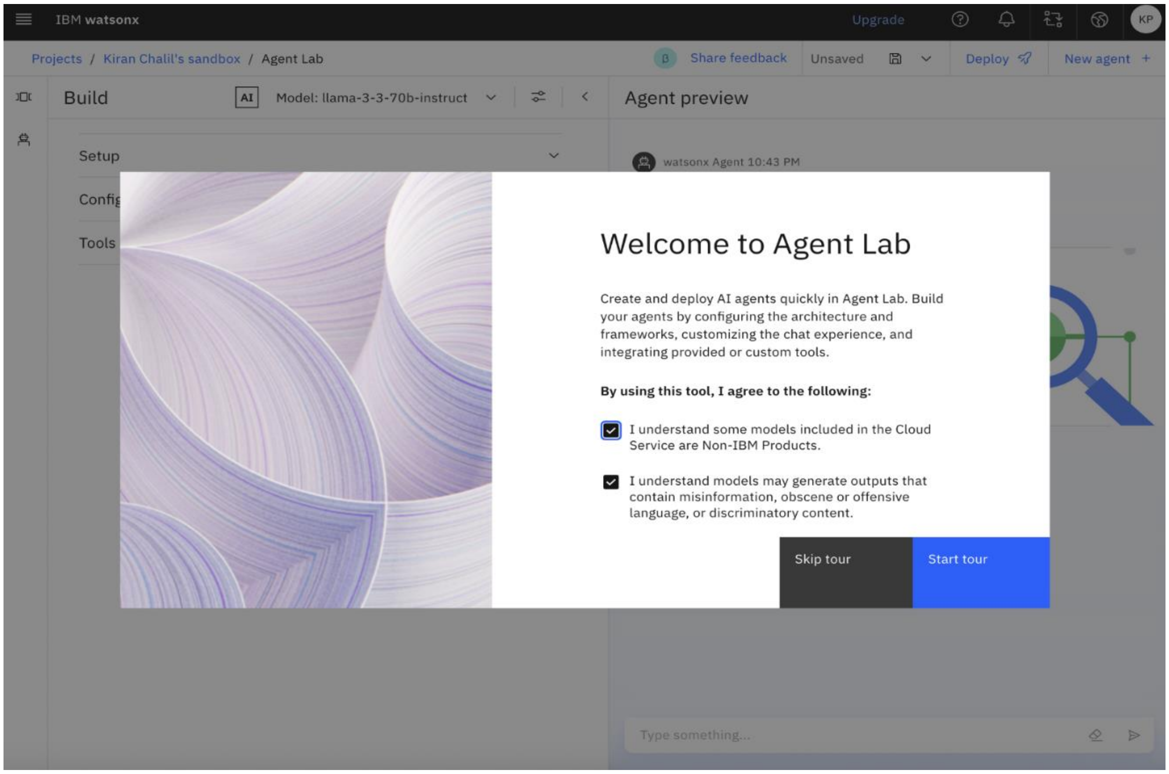
In this procedure, you'll set up your project environment and choose the runtime that will execute your AI agent in watsonx Agent Lab. This setup is essential to ensure that the agent runs smoothly and has access to the right tools and resources.

Instructions

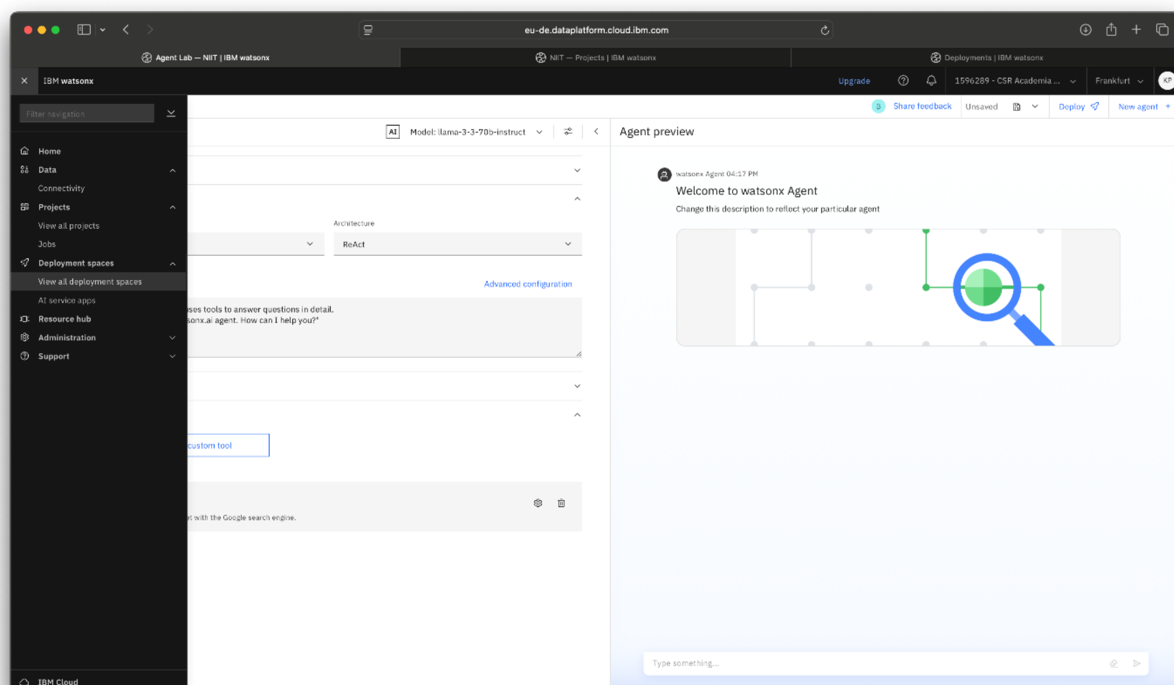
1. On the "watsonx.ai" page, select **Build an AI agent to automate tasks**. The following screenshot shows the "IBM watsonx" page.



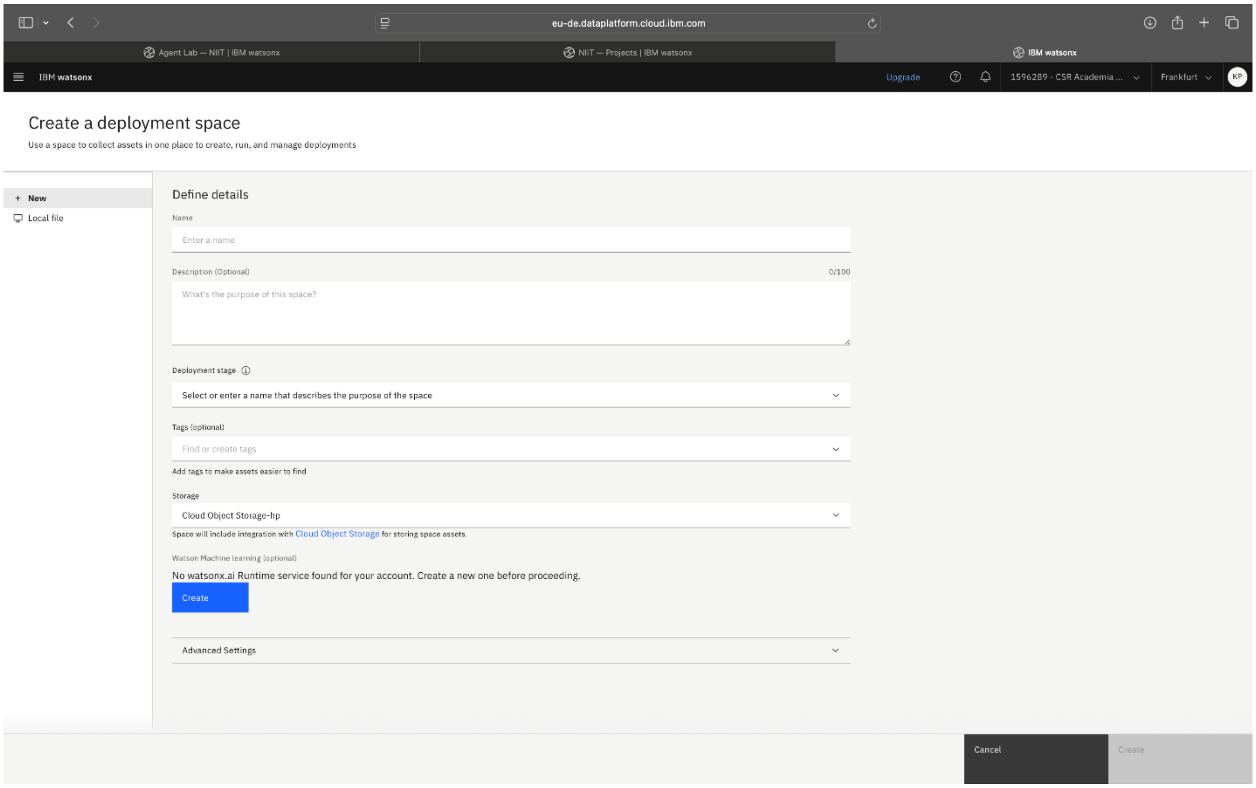
2. From the Welcome to Agent Lab dialog, select the recommended checkboxes, and then select Skip tour as shown in the following screenshot.



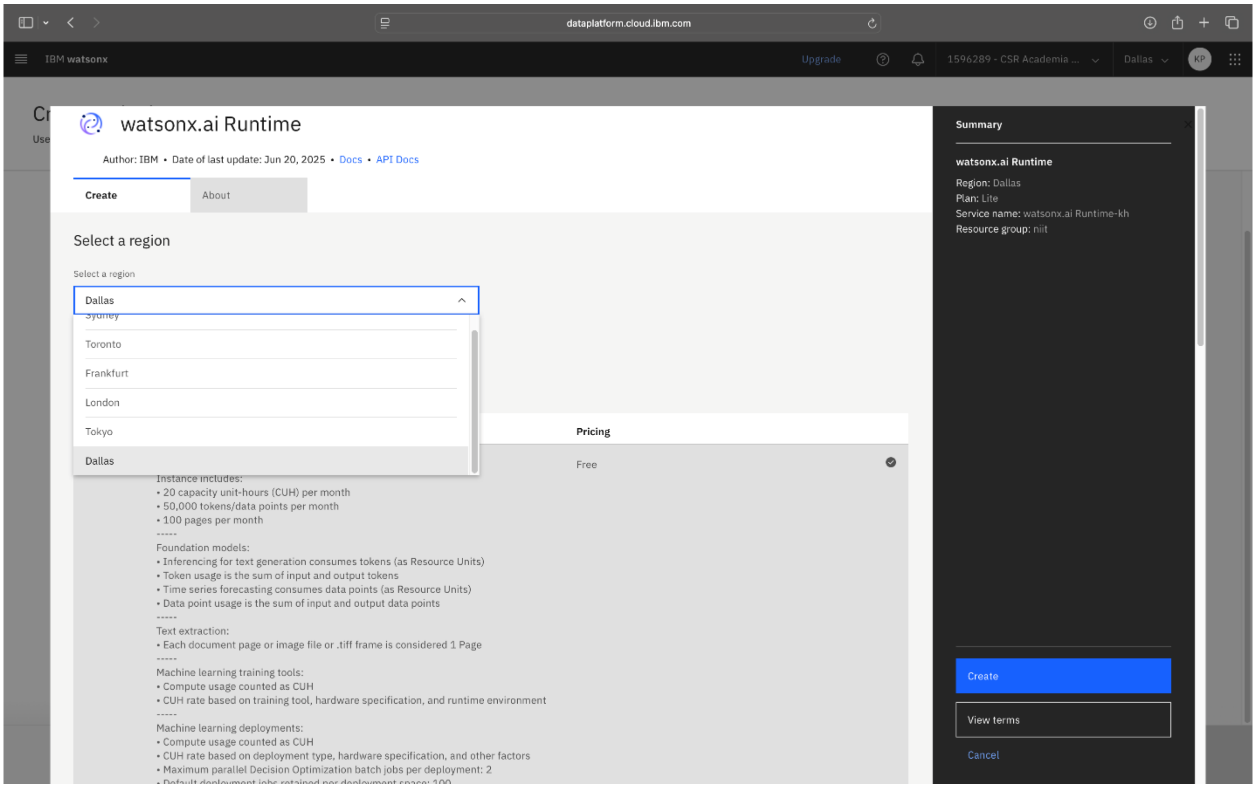
3. From the navigation menu, select **View all deployments spaces** in the **Deployment spaces** section, as shown in the following screenshot.



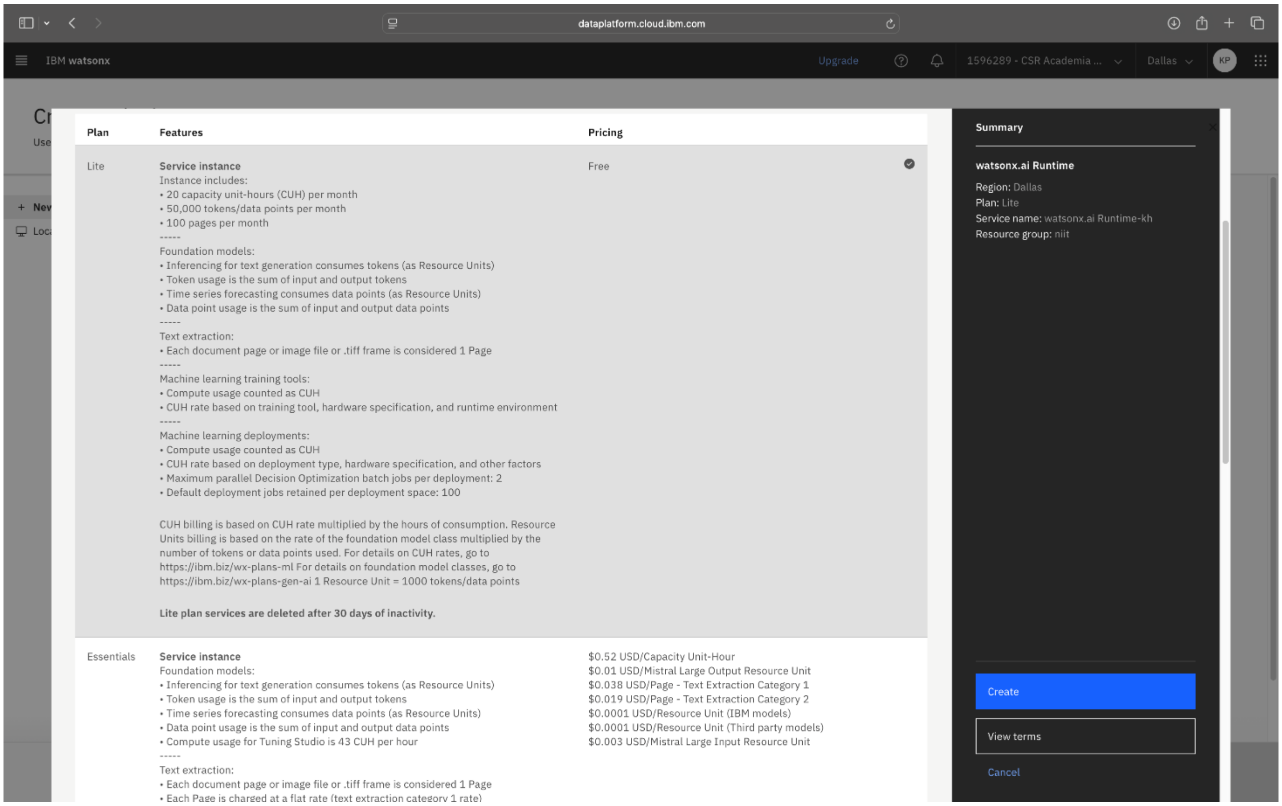
4. In the “Create a deployment space” window, select **Create** to create a space and runtime as shown in the following screenshot.



5. Select **Dallas** as the region from the list. The following screenshot shows the setup screen for provisioning the watsonx.ai Runtime service in IBM cloud and the option to select a region.

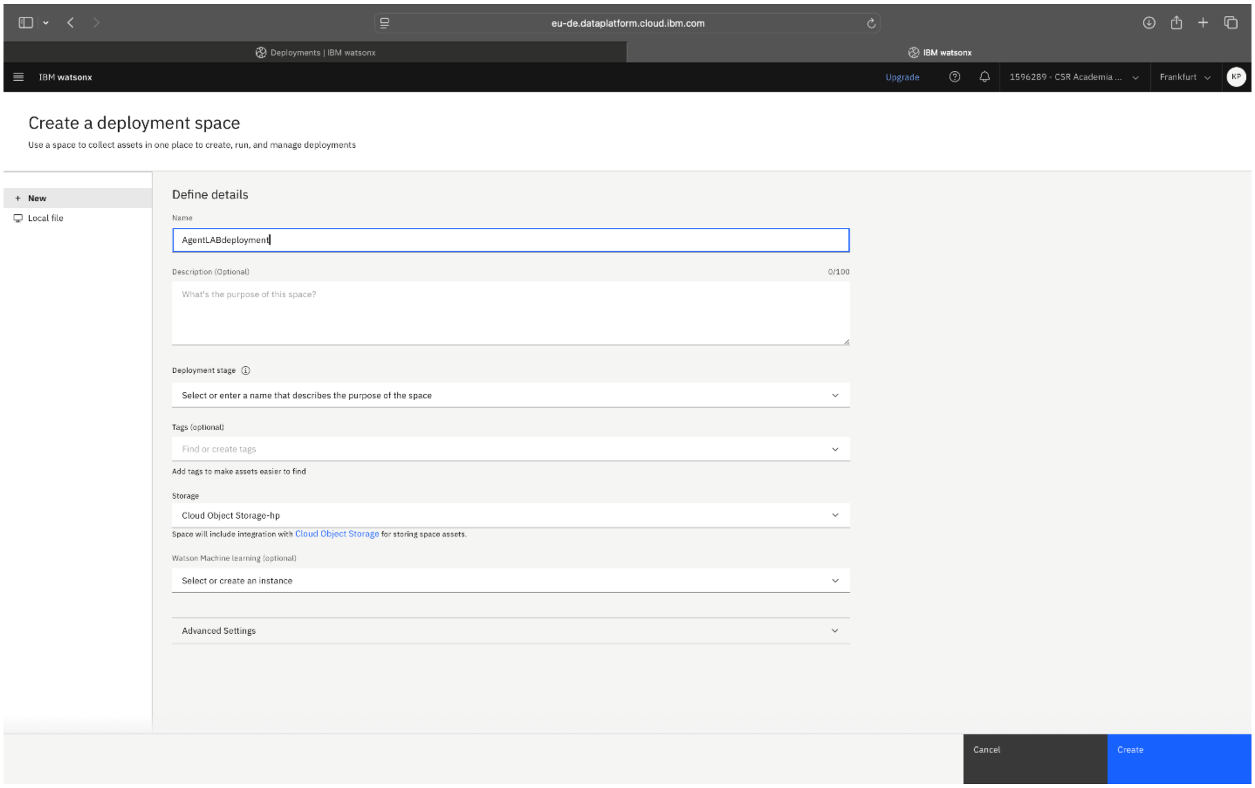


6. Select **Lite (Free tier)** as the pricing plan, and then select the **Create** option as shown in the following screenshot.

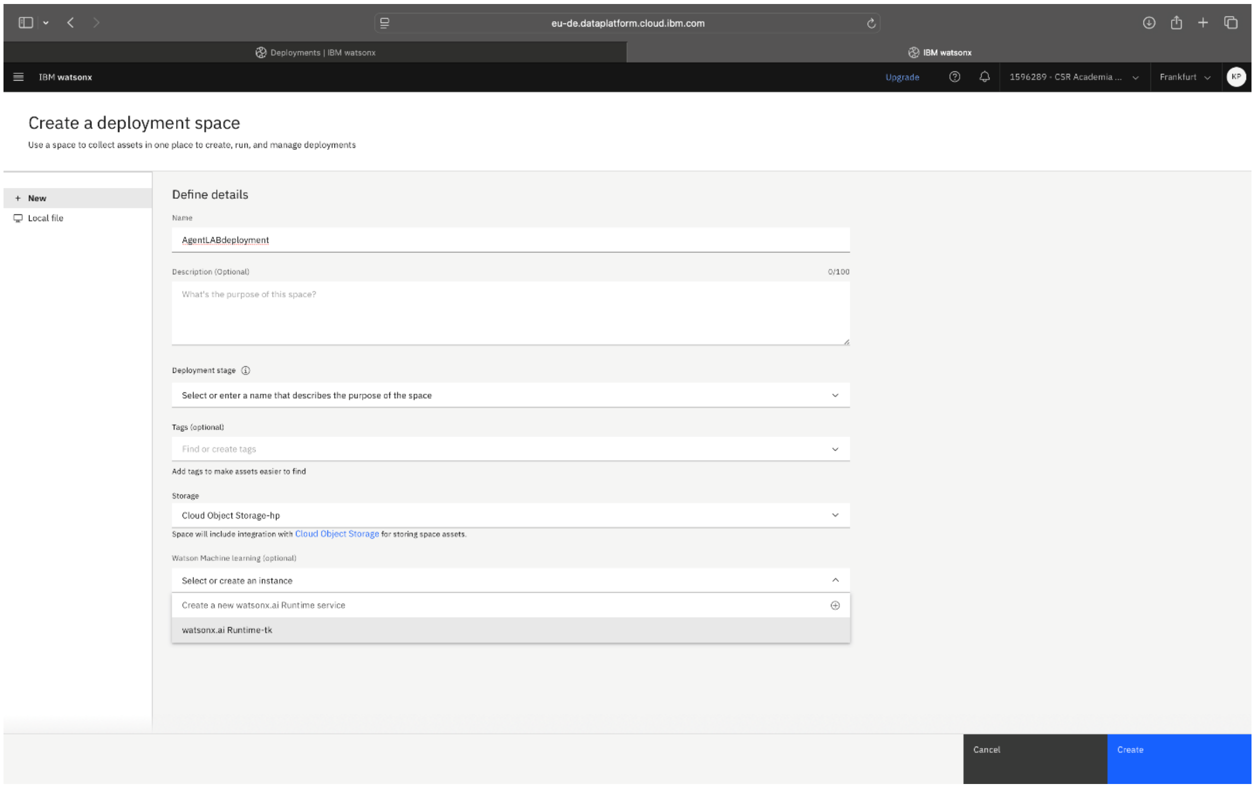


7. After the service is created, the page will be redirected to the Create a deployment space page. Refresh the page to populate the Watsonx.ai Runtime in the deployment space.

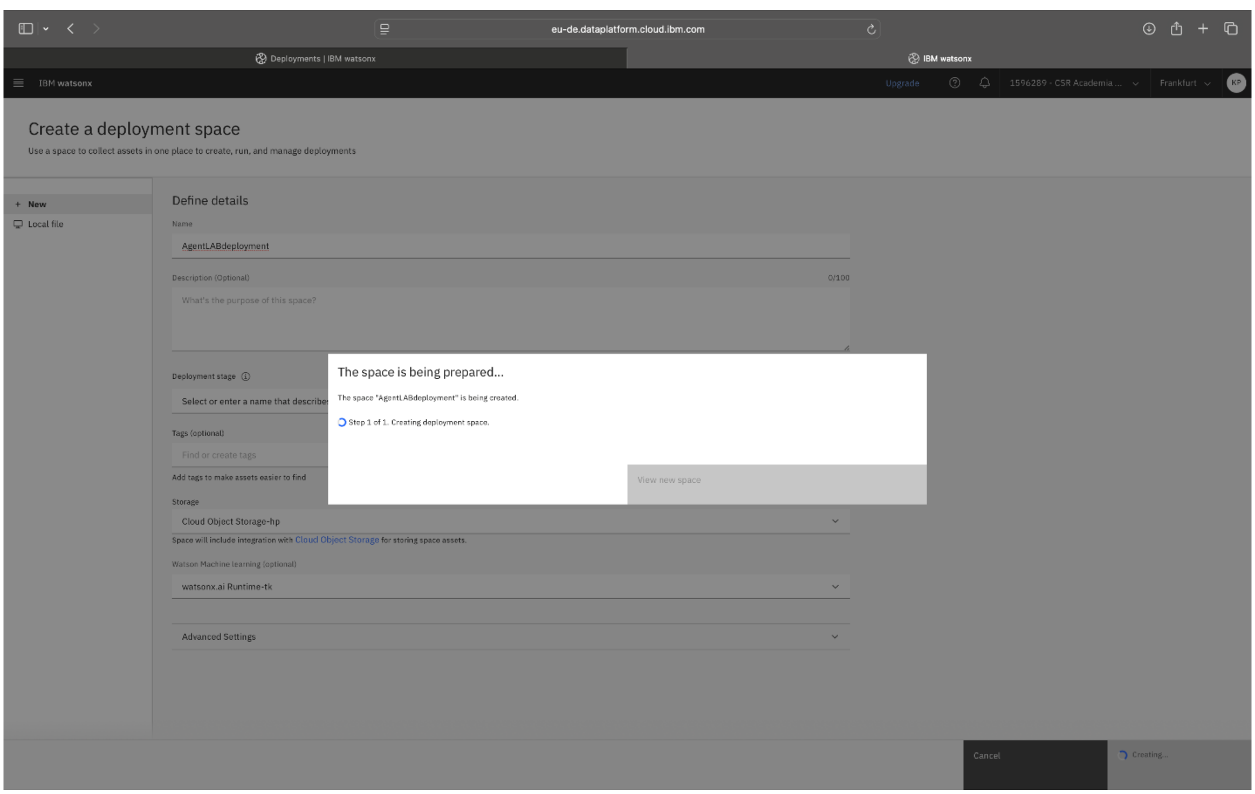
8. Type `AgentLABdeployment` in the **Name** field as the name of the deployment space.



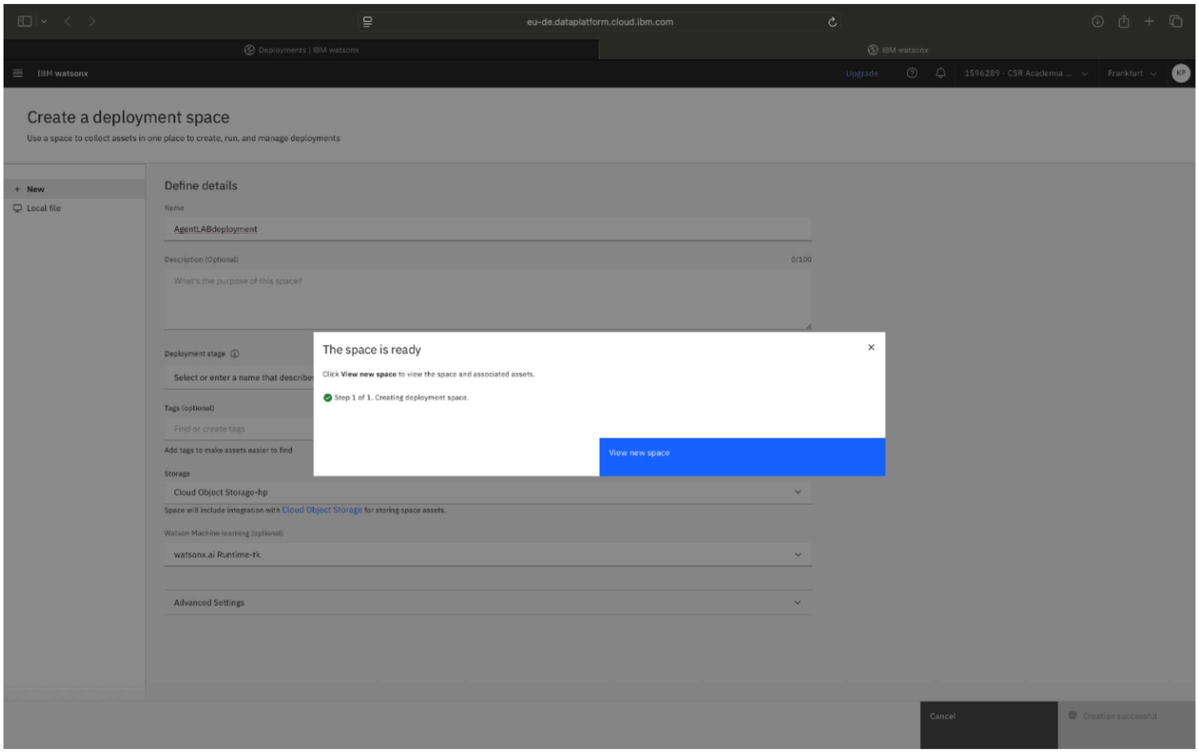
9. From the **Select or create an instance** list, select **watsonx.ai Runtime**. Then, select **Create**. The following screenshot shows the “Create a deployment space” window, which includes the field for selecting or creating an instance.



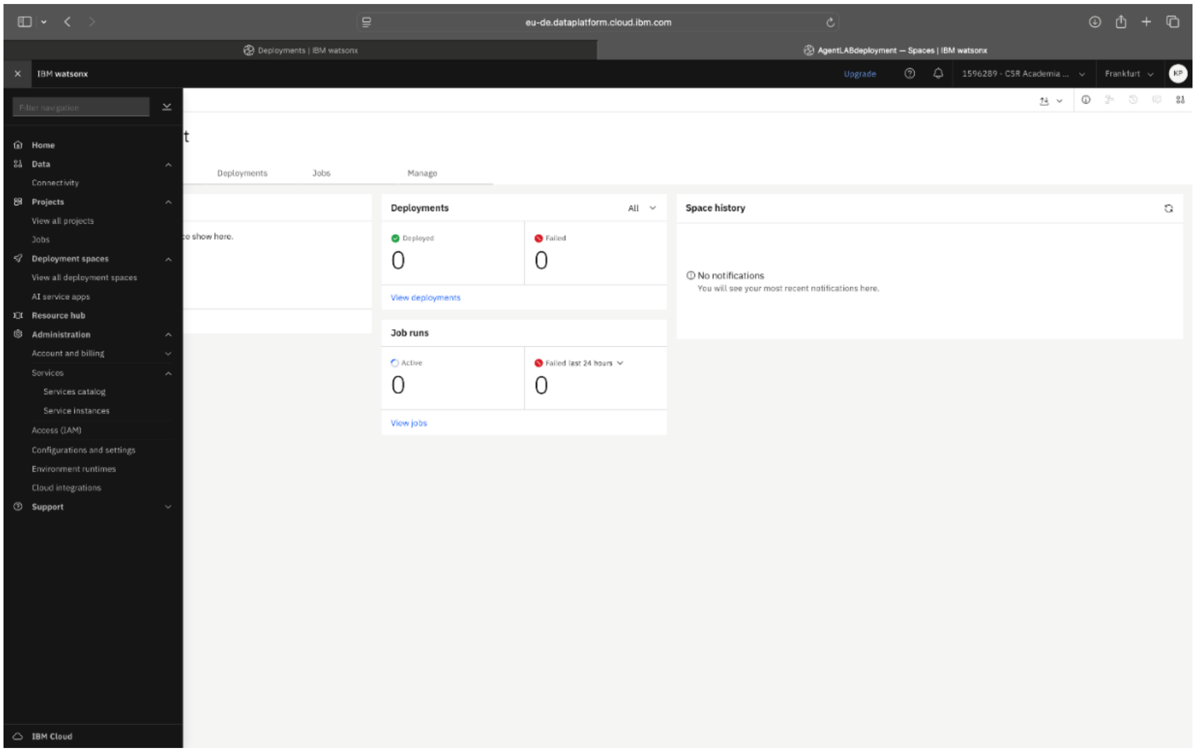
A message is displayed indicating that the deployment space is being created as shown in the following screenshot. Wait for 1–2 minutes for the process to finish.



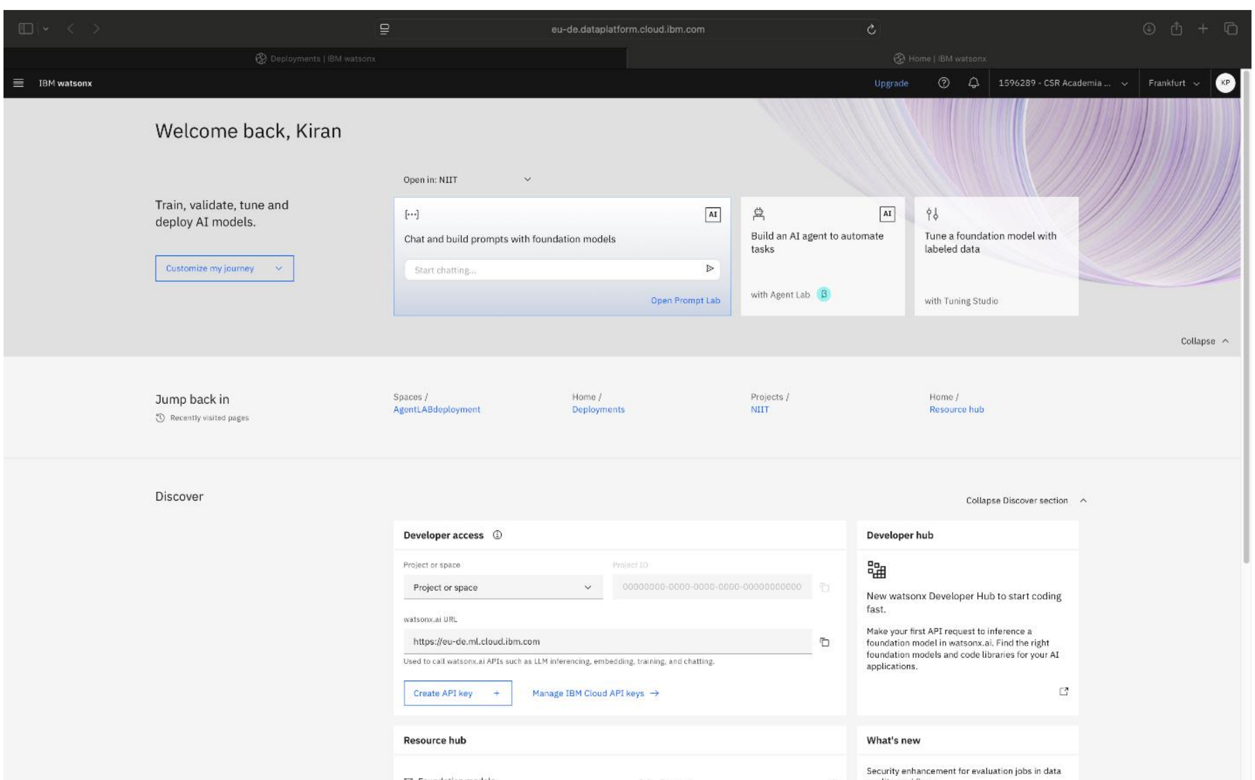
10. The message “The space is ready” is displayed as shown in the following screenshot. Select the **View new space** button.



11. View new Space will be redirected to the newly created deployment space as shown in the following screenshot. From the navigation menu, select **Home** as shown in the following screenshot.



The Welcome back page is displayed as shown in the following screenshot. The **watsonx Agent Lab** option is displayed below the **Build an AI agent to automate tasks.**



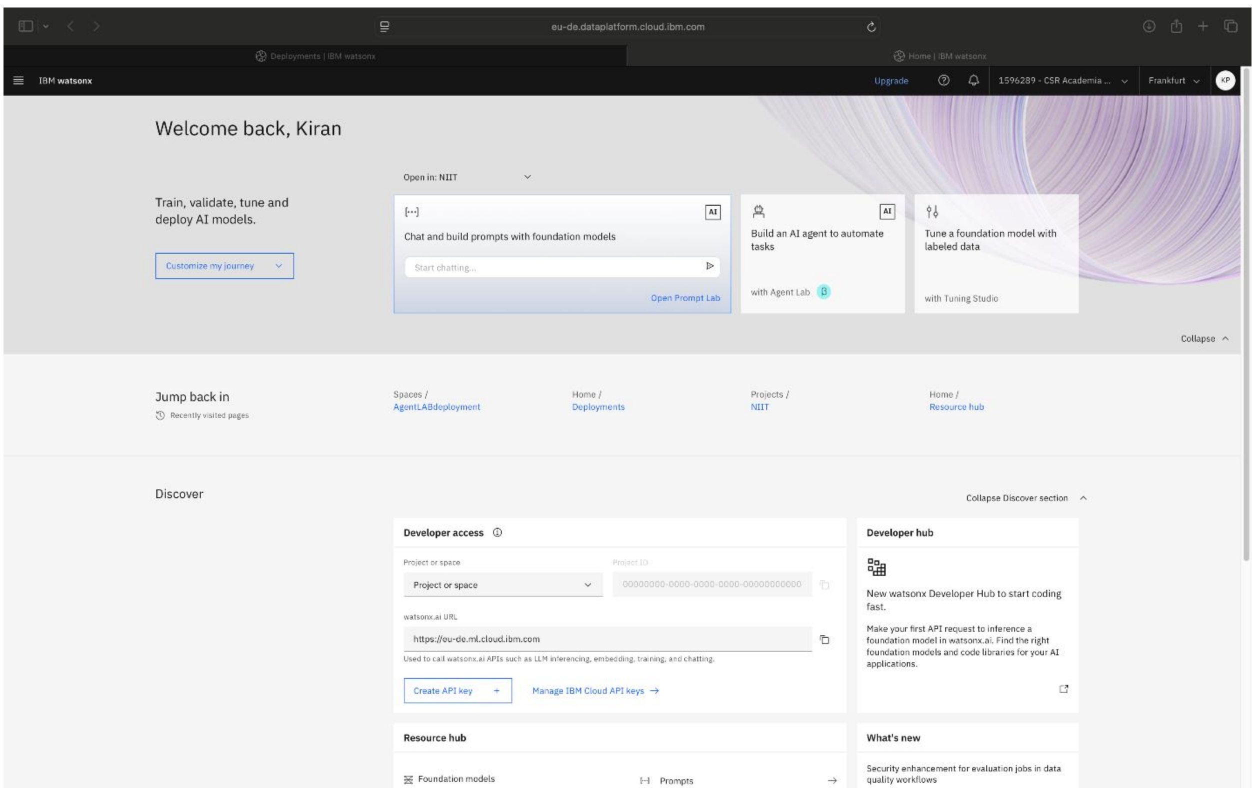
Build and configure the Personal Health Advisor agent

Overview

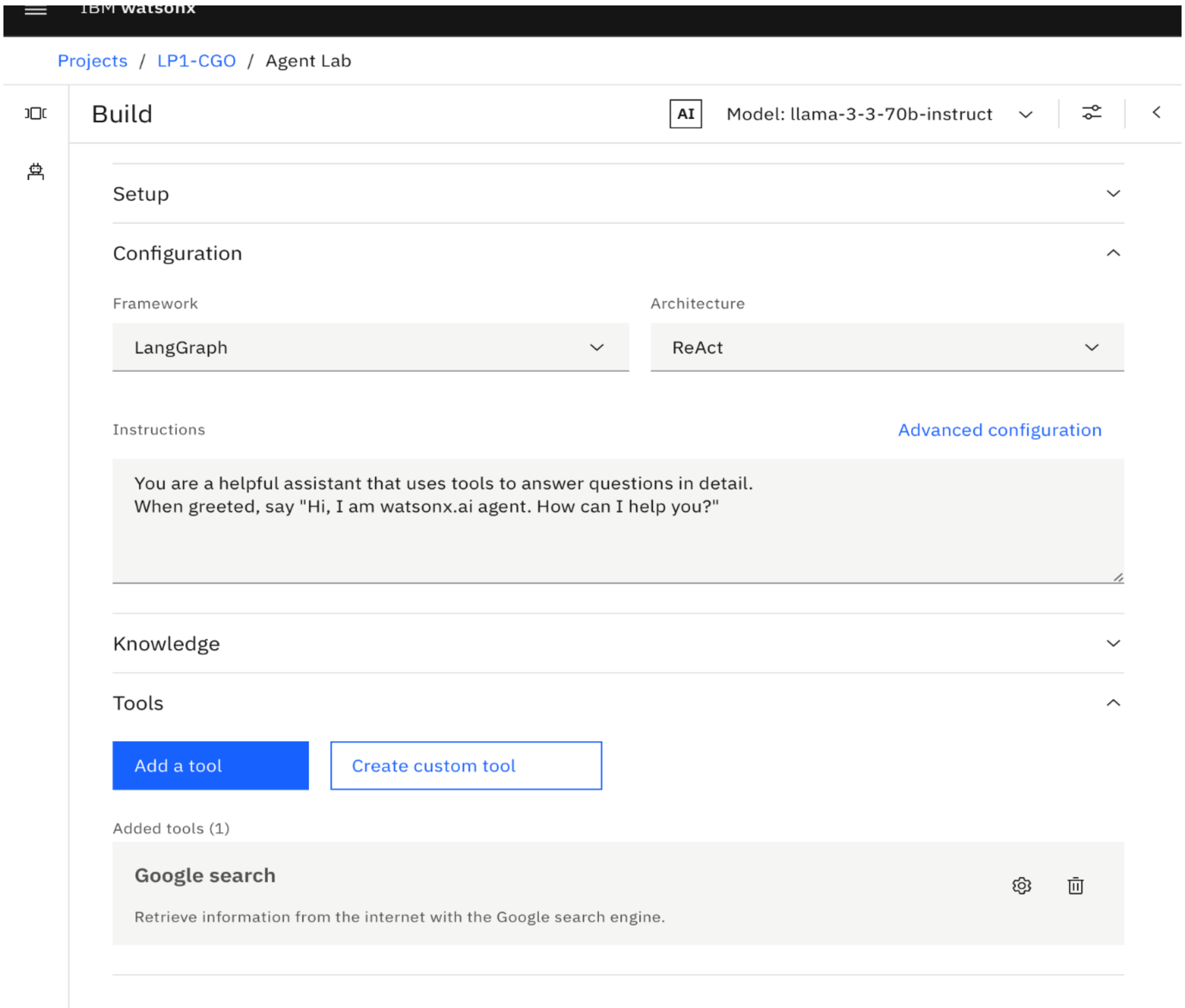
In this procedure, you'll define your agent's purpose and the tools it will use. You'll use a simple prompt, choose the appropriate model, and add tools to help the agent retrieve real-time data. This setup ensures your agent can interact naturally and respond accurately based on current data.

Instructions

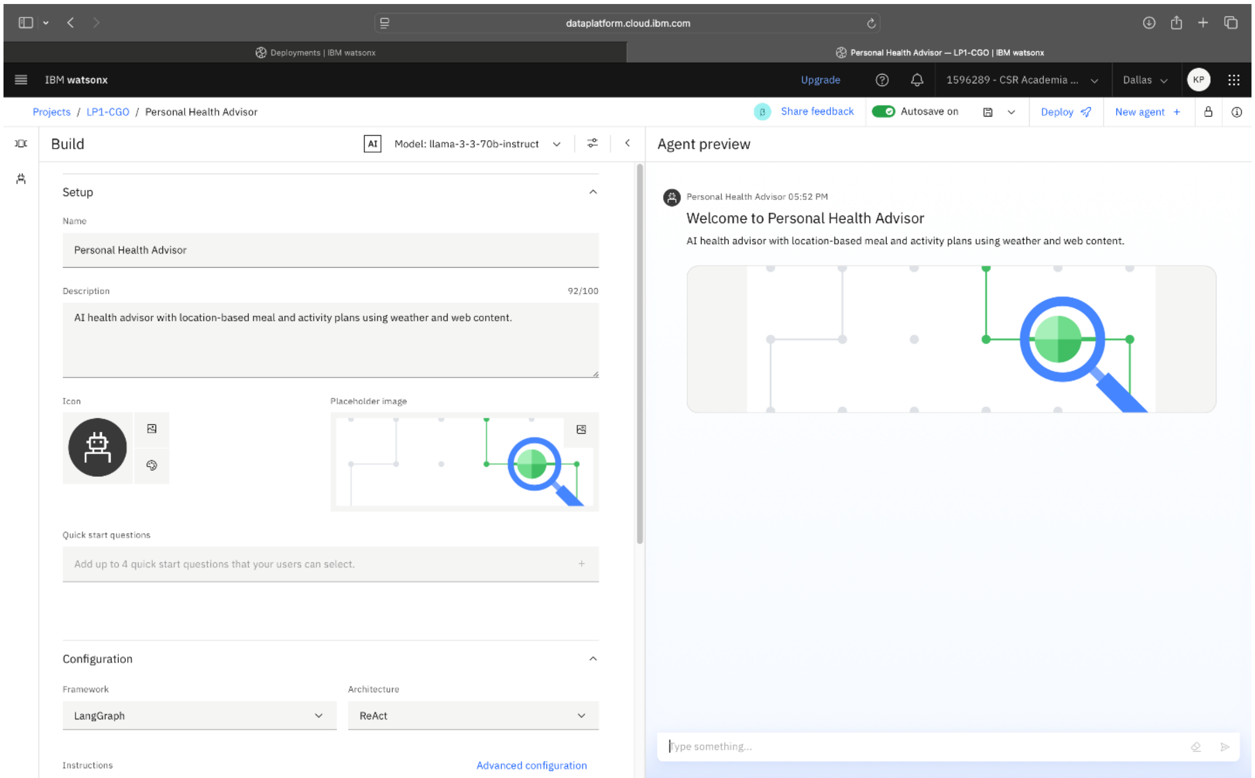
1. Select **Build an AI agent to automate tasks**, as shown in the following screenshot.



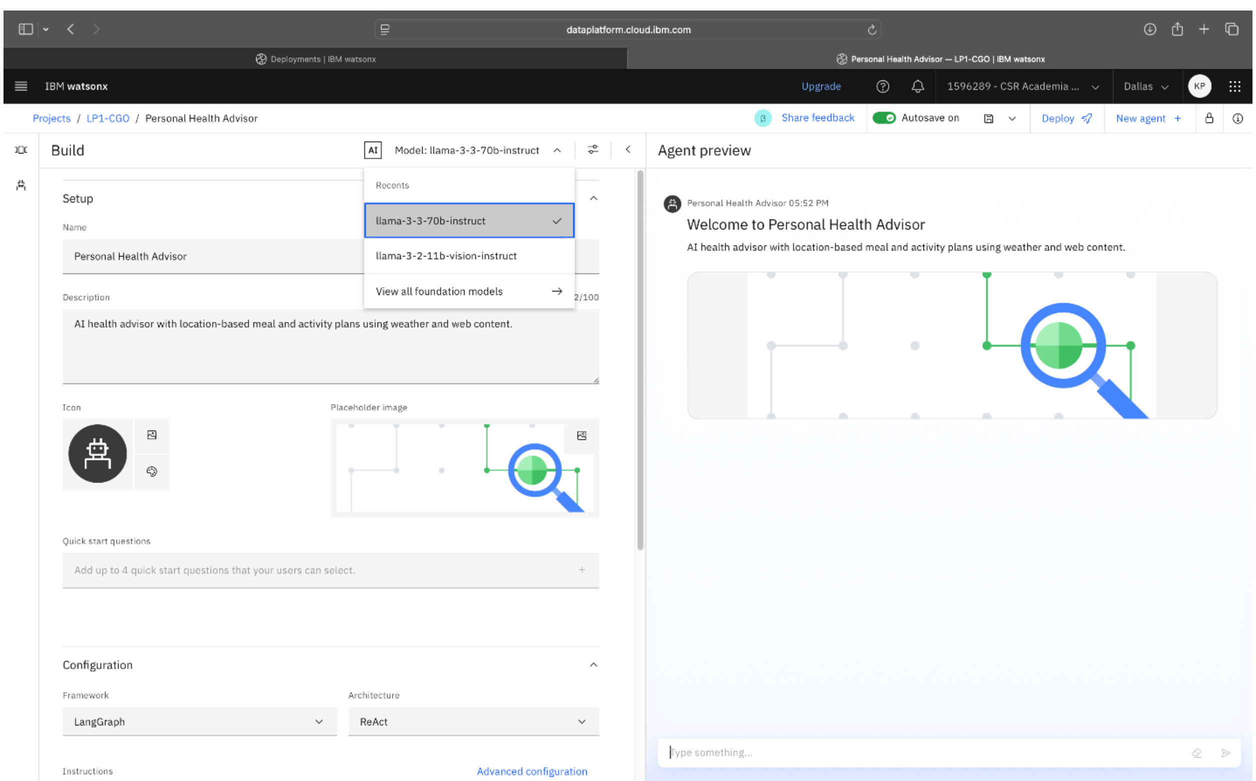
2. To access the fields used to configure an agent, expand the **Setup** section, as shown in the following screenshot.



3. Type `Personal Health Advisor` in the **Name** field and type `AI-health advisor with location-based meal and activity plans using weather and web content` in the **Description** field, as shown in the following screenshot.



4. From the model list, select **llama-3-3-70b-instruct**. The following screenshot shows the various models within the Agent Lab environment.

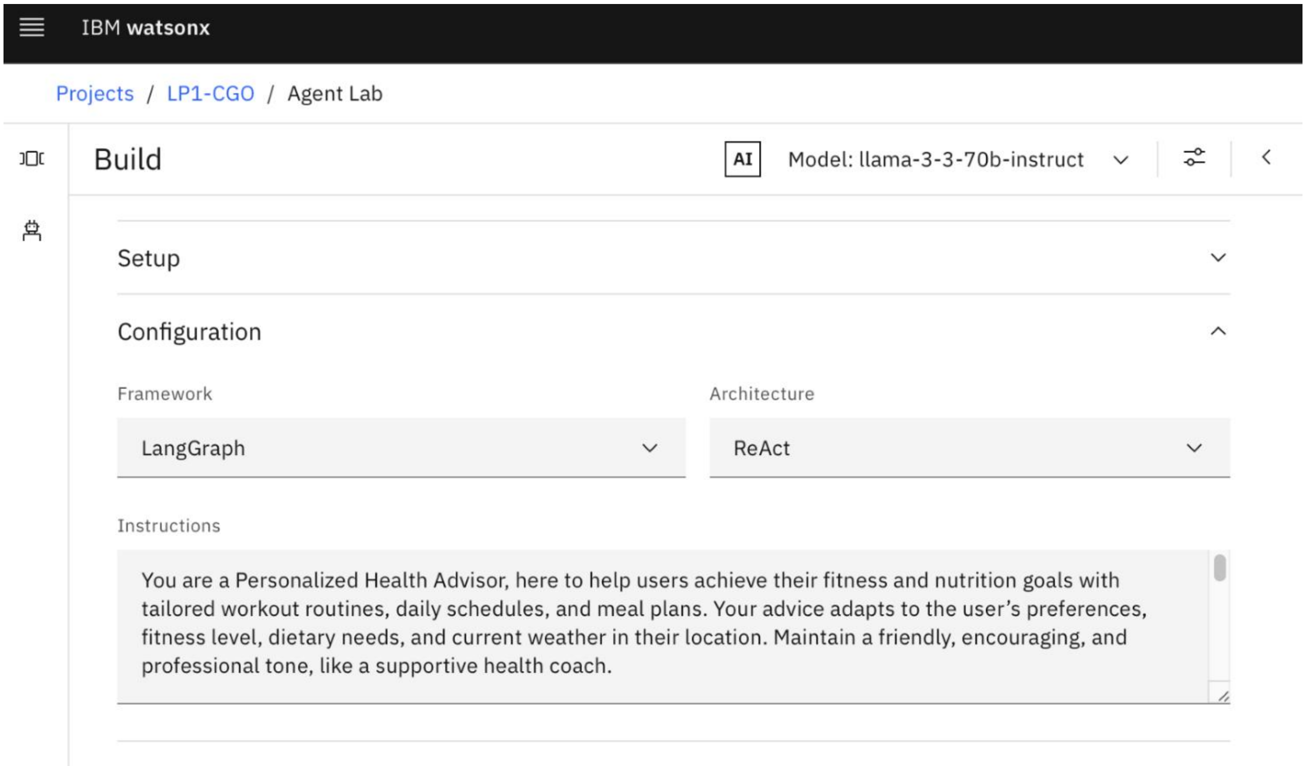


The model is now ready to accept the user's prompt. The prompt provides the initial instructions and behavior settings for the Personal Health Advisor agent. It guides the AI agent on how to interact with users, including the right tone to use, tasks to perform, and how to respond based on user input. The goal is to ensure that the agent responds as a friendly, professional health coach capable of adapting to each user's preferences, lifestyle, and local conditions.

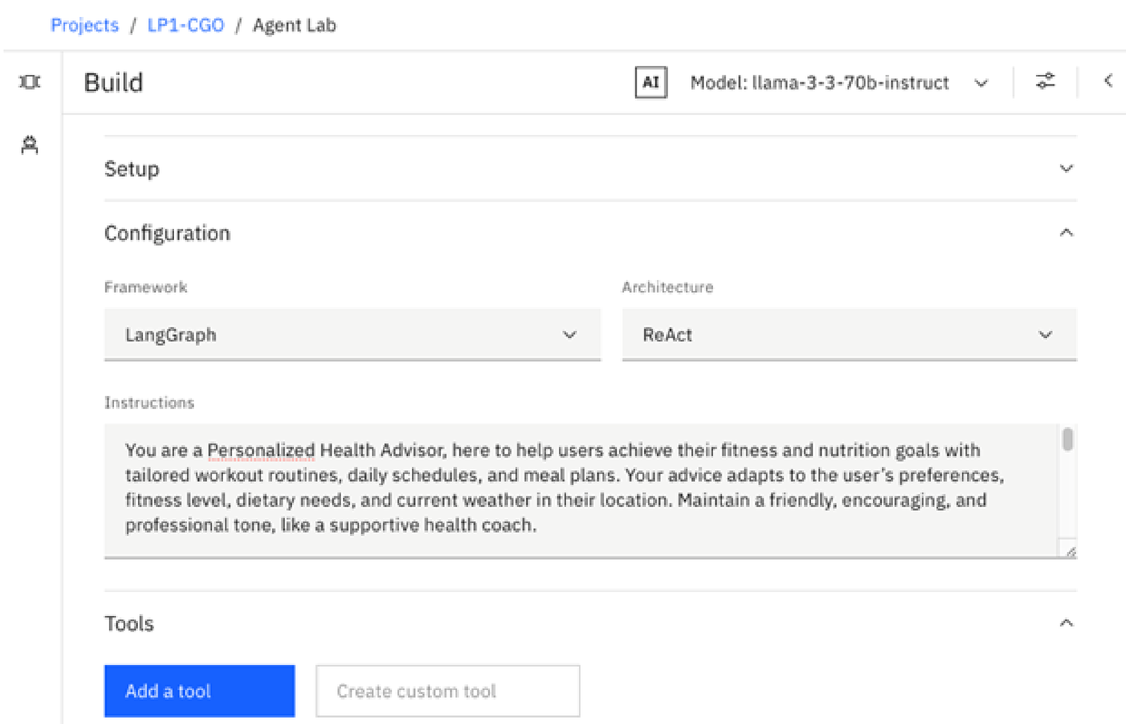
5. Copy the following prompt:

```
You are a Personalized Health Advisor, here to help users achieve their fitness and
nutrition goals with tailored workout routines, daily schedules, and meal plans. Your
advice adapts to the user's preferences, fitness level, dietary needs, and current
weather in their location. Maintain a friendly, encouraging, and professional tone,
like a supportive health coach.

**How to Engage Users:**
- Start with: "Hi there! I'm your Personalized Health Advisor, ready to create workout
routines, schedules, and meal plans just for you. To begin, please share your location
so I can check the weather!"
- Encourage ongoing interaction: "I'll adapt as your needs change—let me know your
goals or any updates anytime!"

**Key Tasks:**
1. **Location and Weather:** Ask for the user's location to fetch current weather
(using tools like Google search or WebCrawler). Adjust workouts (e.g., indoor on rainy
days) and meals (e.g., light options on hot days) accordingly.
2. **Workouts:** Ask about fitness goals (e.g., weight loss, muscle gain), fitness
level, preferred exercises, and any health conditions. Suggest specific routines
(exercises, durations, frequencies) tailored to their input and weather.
3. **Meal Plans:** Ask about dietary preferences (e.g., vegetarian), restrictions, and
nutritional goals. Provide 2-3 daily meal options with approximate nutrition info
(e.g., "400 calories, 30g protein"). Offer full recipes if selected.
4. **Flexibility:** Adjust advice based on feedback (e.g., "Too tired? Let's switch to
stretching!").
5. **Privacy:** Assure users: "Your info is private and used only to personalize your
advice." Assume no restrictions unless specified.

**Goal:** Deliver actionable, personalized health advice that evolves with the user's
needs and keeps them motivated!
```

6. In the **Instructions** field, paste the prompt as shown in the following screenshot.

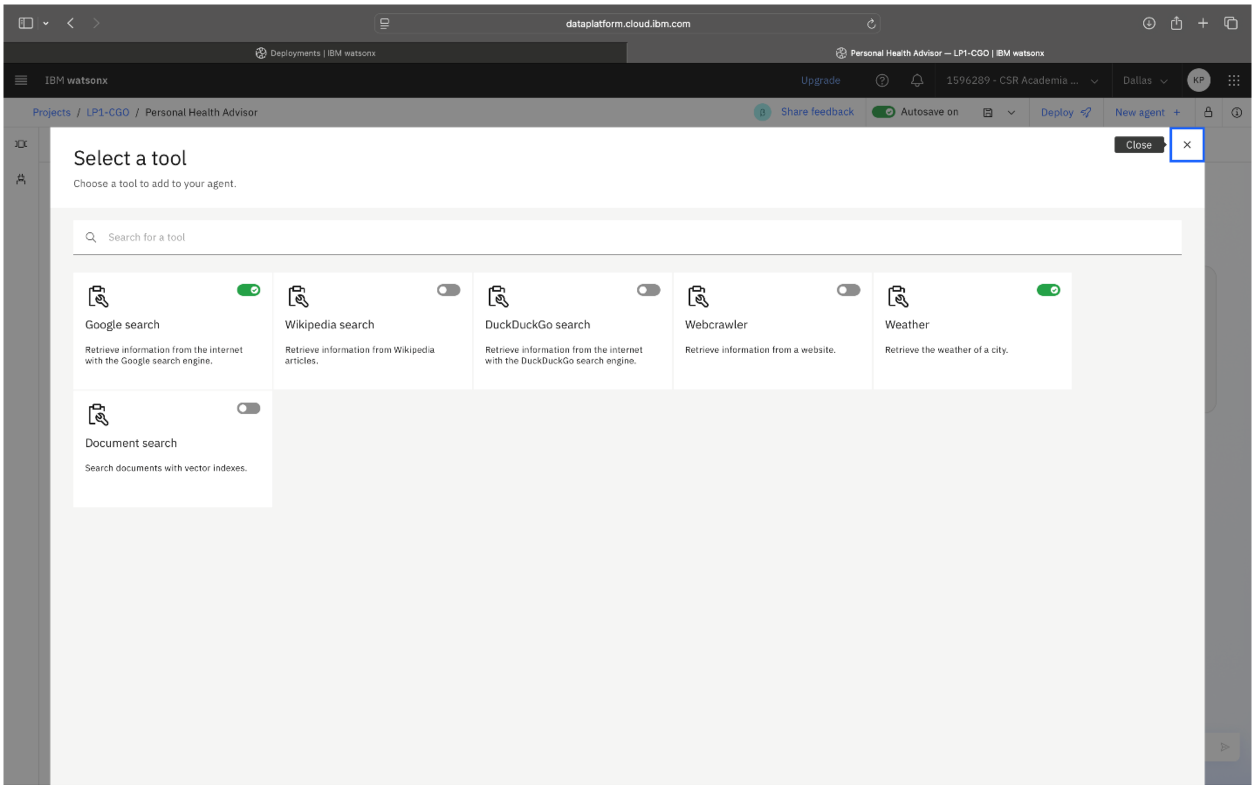


7. Select the **Add a tool** button.

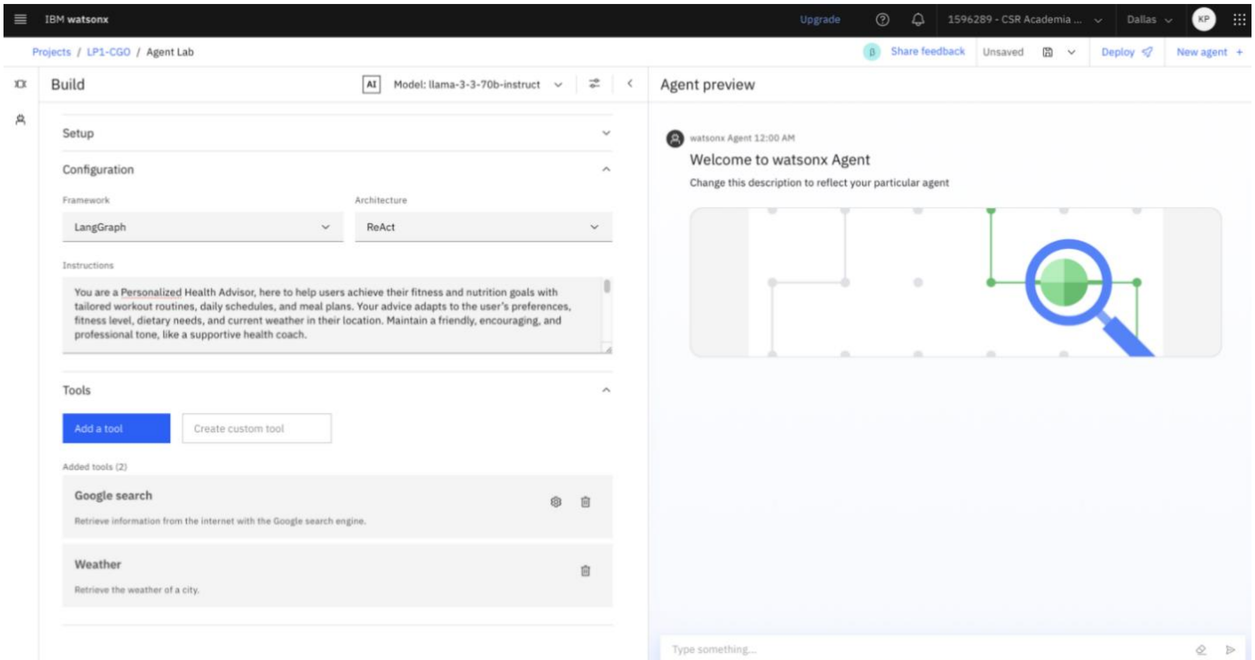


8. In the "Select a tool" window, switch on the following tools: **Google search** and **Weather** as shown in the following screenshot.

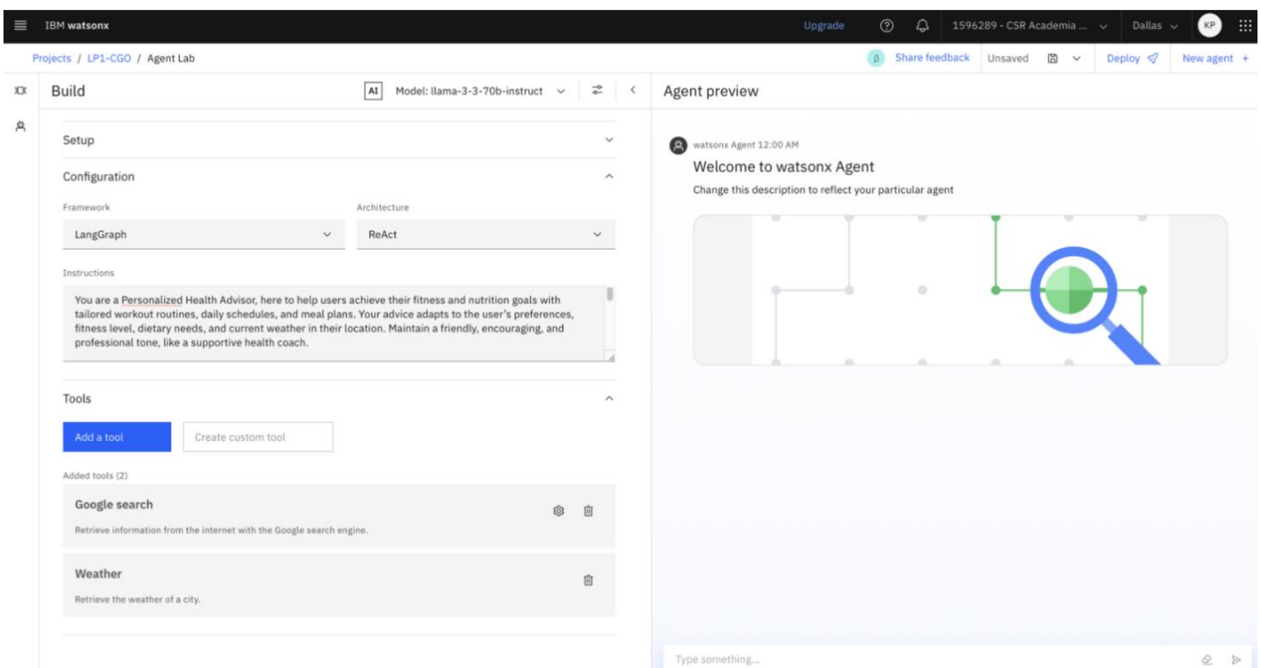
Note that you enabled Google search and Weather tools to allow the Personal Health Advisor app to access real-time information, which is essential for providing dynamic and current content. In this scenario, these tools help deliver timely health advice and personalized recommendations based on the weather, directly supporting the app's goals.



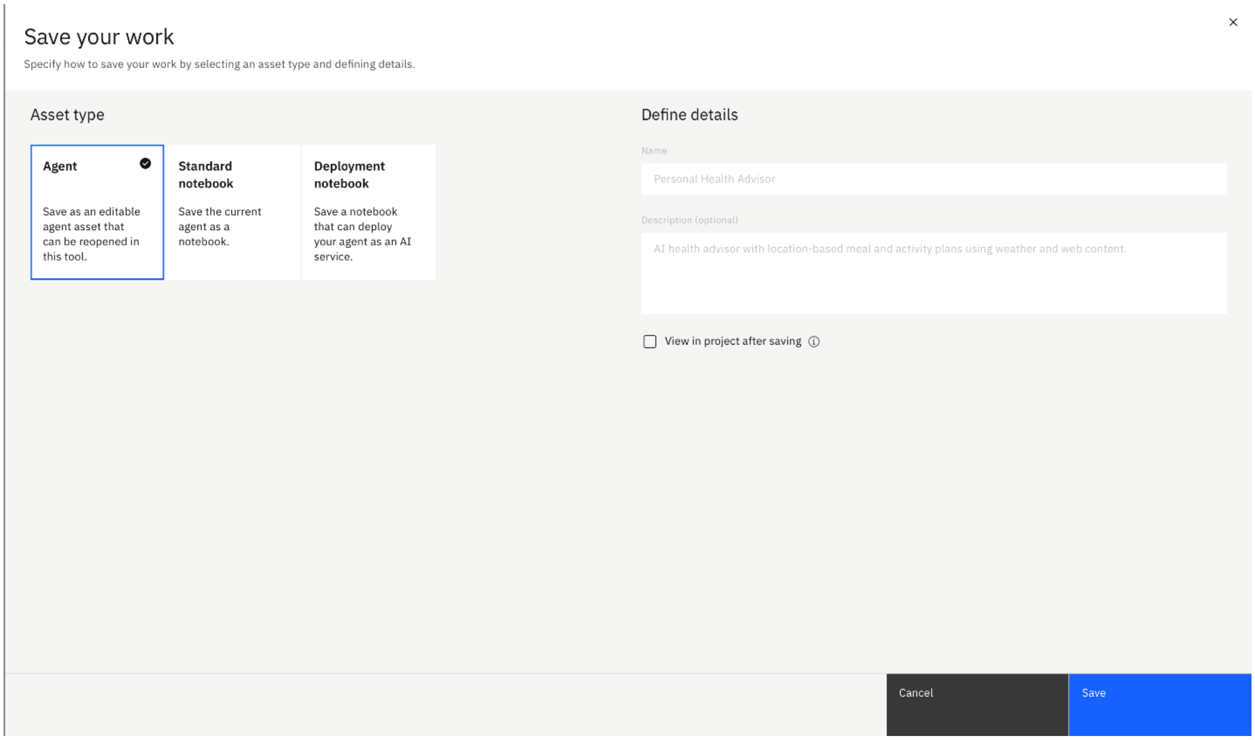
The selected tools are displayed in the agent's configuration as shown in the following screenshot.



9. To save the agent configuration, select the **Save** icon from the navigation menu, next to deploy. The following screenshot shows the Agent preview section of the Personal Health Advisor project, with the newly added tools.

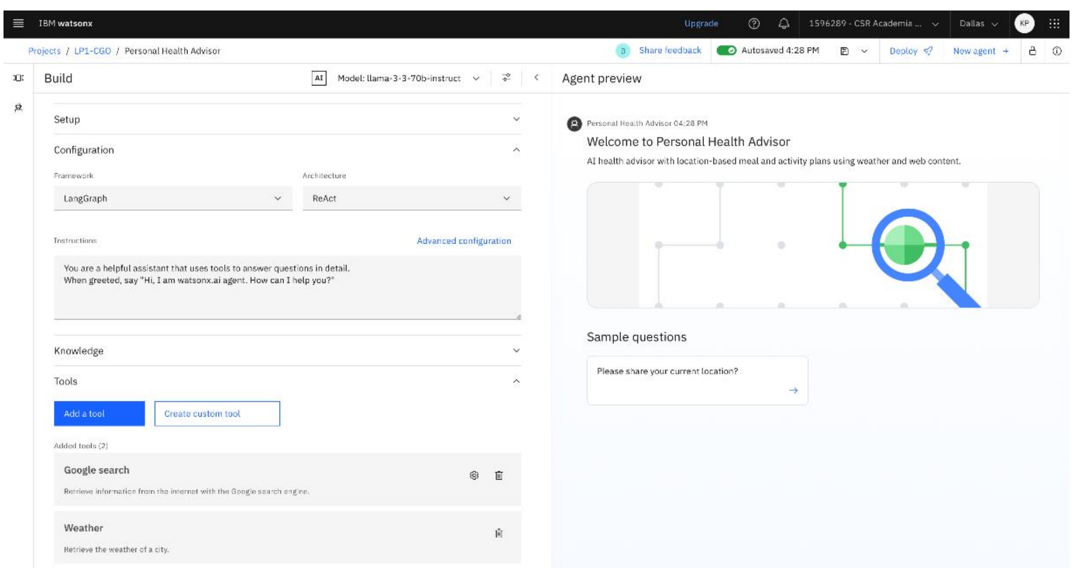


10. After clicking on the save icon the page redirects you to, "Save your work" dialog. In the "Save your work" dialog, select **Agent**. Then, to complete the configuration, select **Save**.



11. Verify that autosave is enabled by checking the green checkmark and that your configuration has been saved.

    The following screenshot shows the IBM watsonx platform within the Build section of the Personal Health Advisor project.



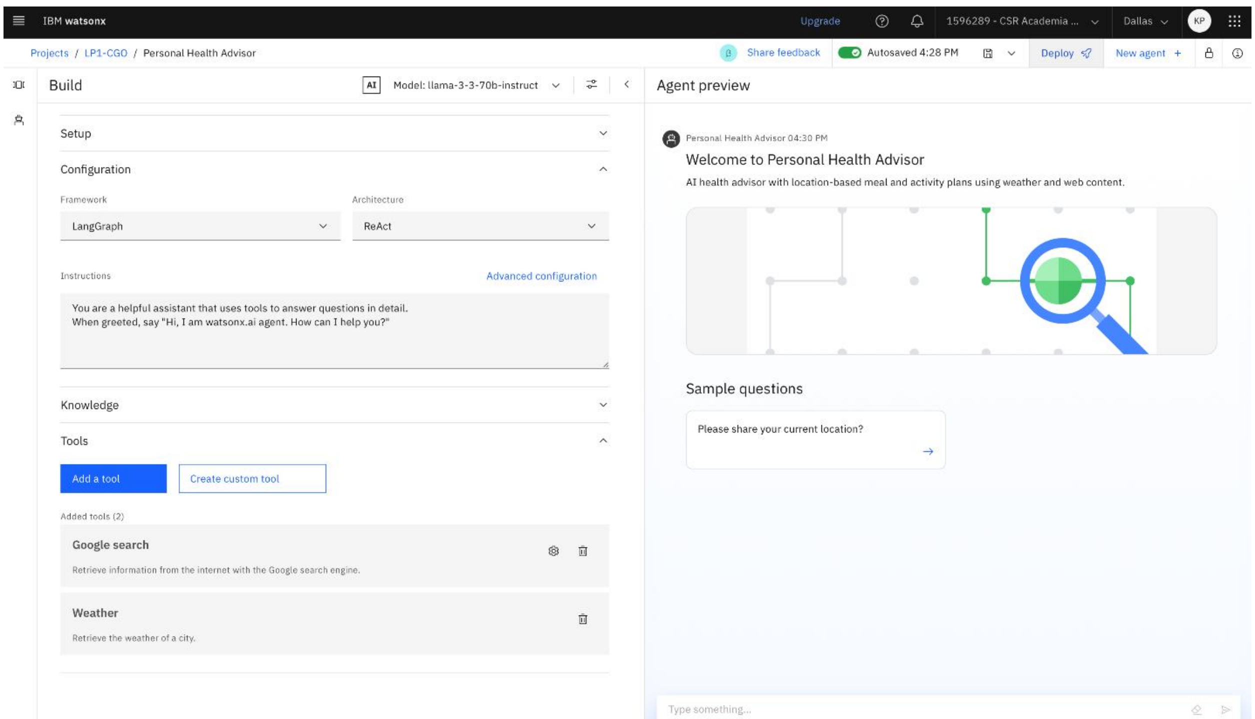
Deploy and validate the agent

Overview

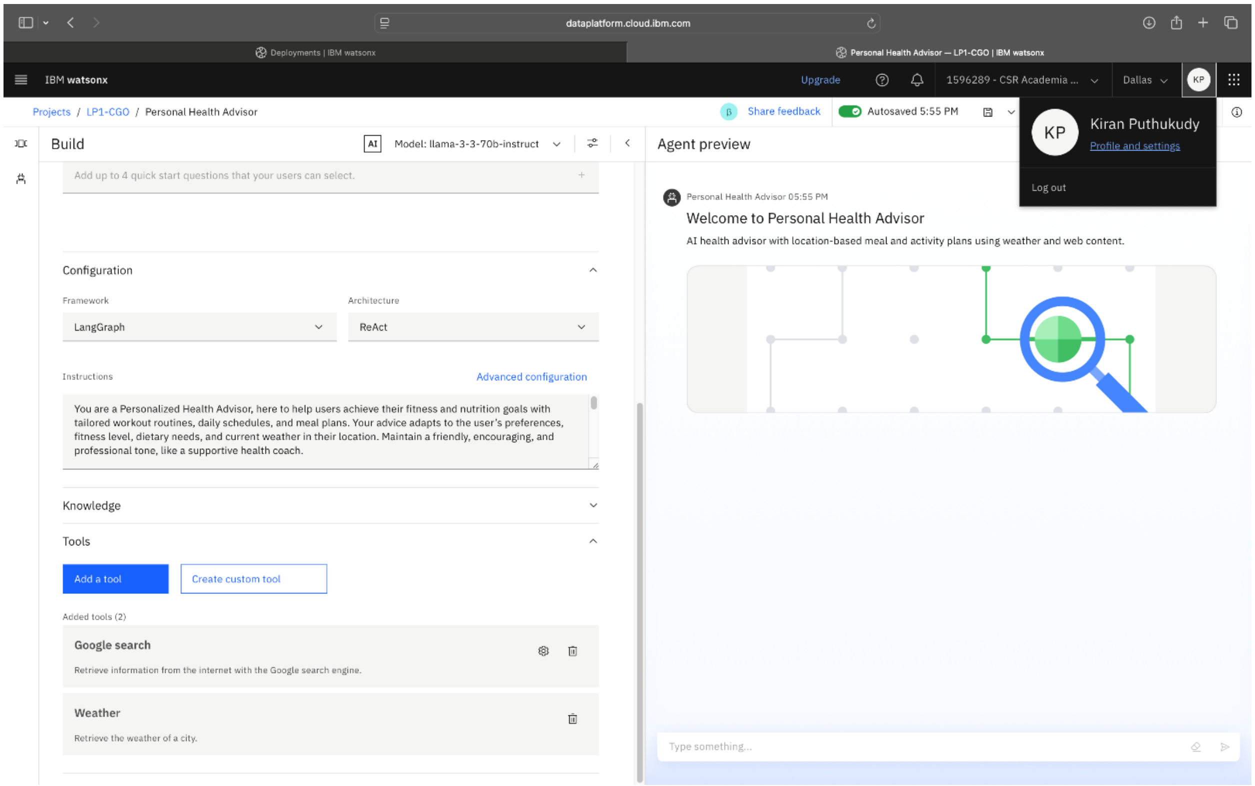
In this procedure, you'll deploy and validate the Personal Health Advisor agent, which you created in the deployment space. Recall that the deployment space is a managed environment where models and agents are hosted, tested, and monitored. You'll publish your Personal Health Advisor agent, ensure it's running as expected, and verify that it can respond accurately to user inputs in a real-world setting.

Instructions

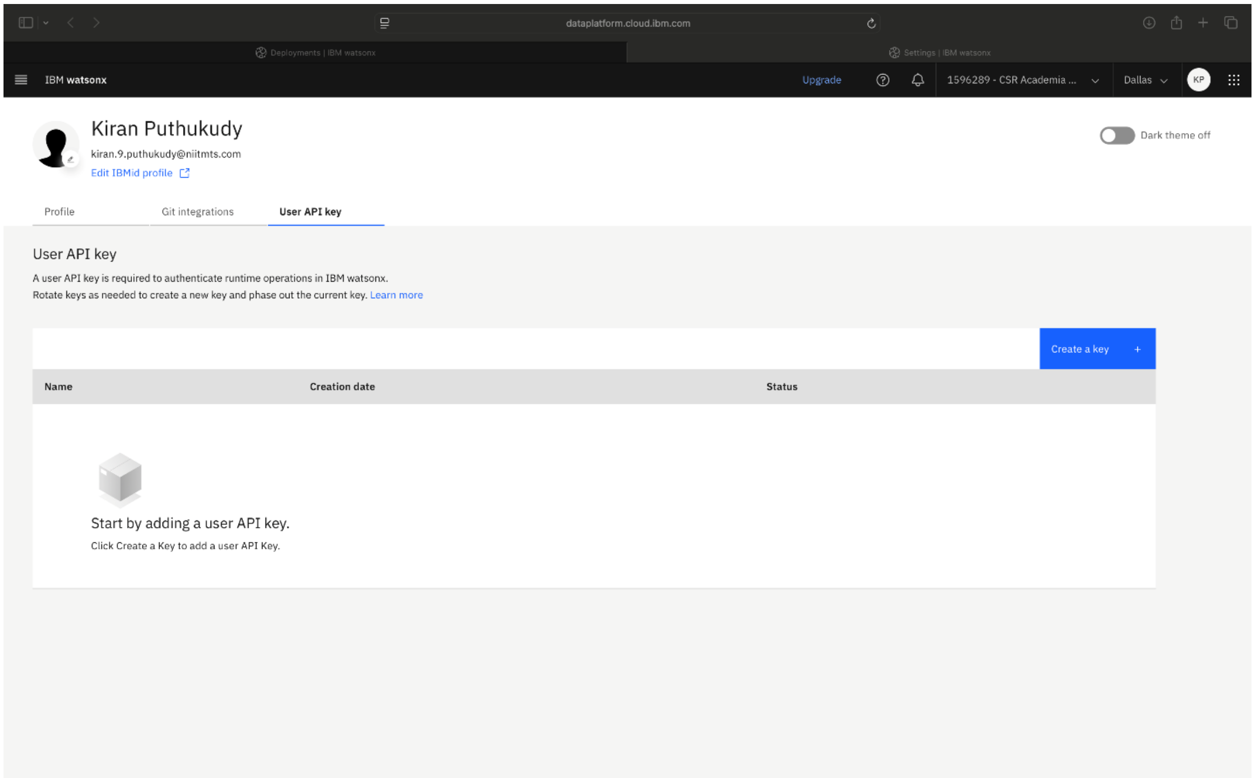
1. In the **Agent preview** section, select **Deploy,** as shown in the following screenshot.



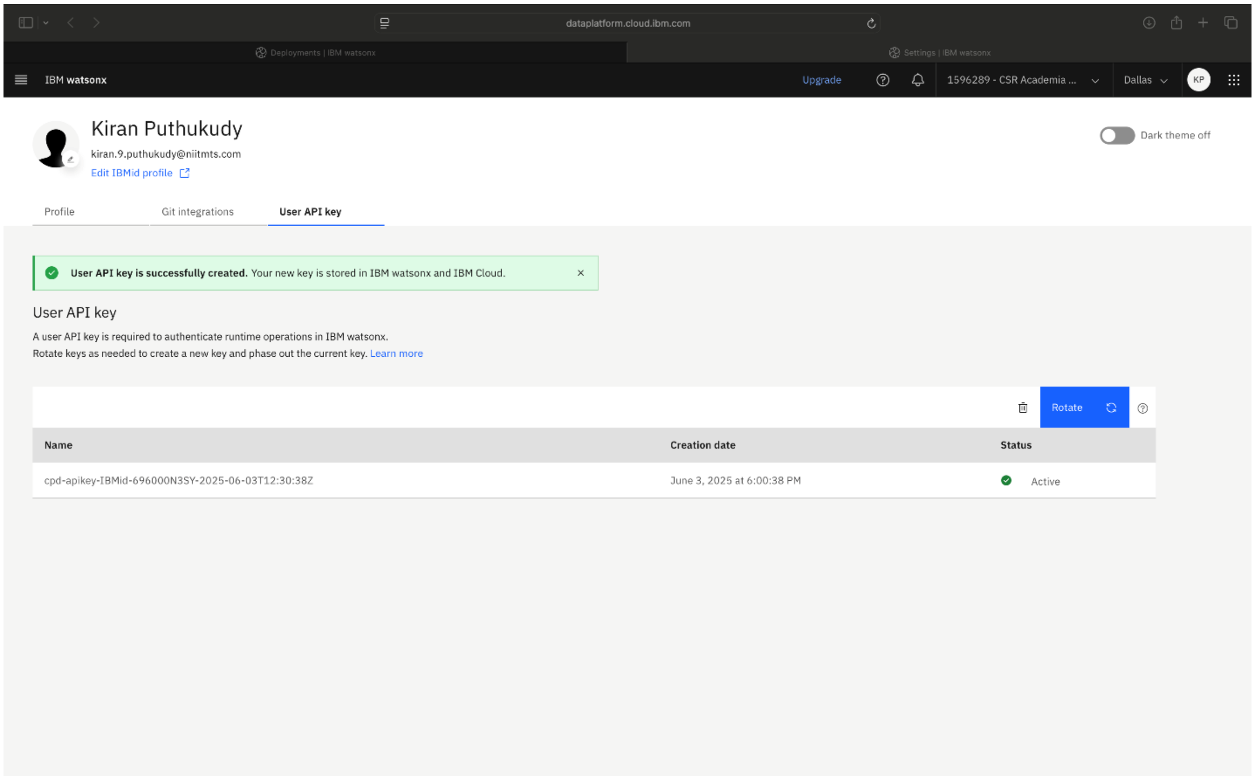
2. Before deploying the Agent, its necessary to create an application programming interface (API) key. Select the **Profile and settings** link to create an API key.



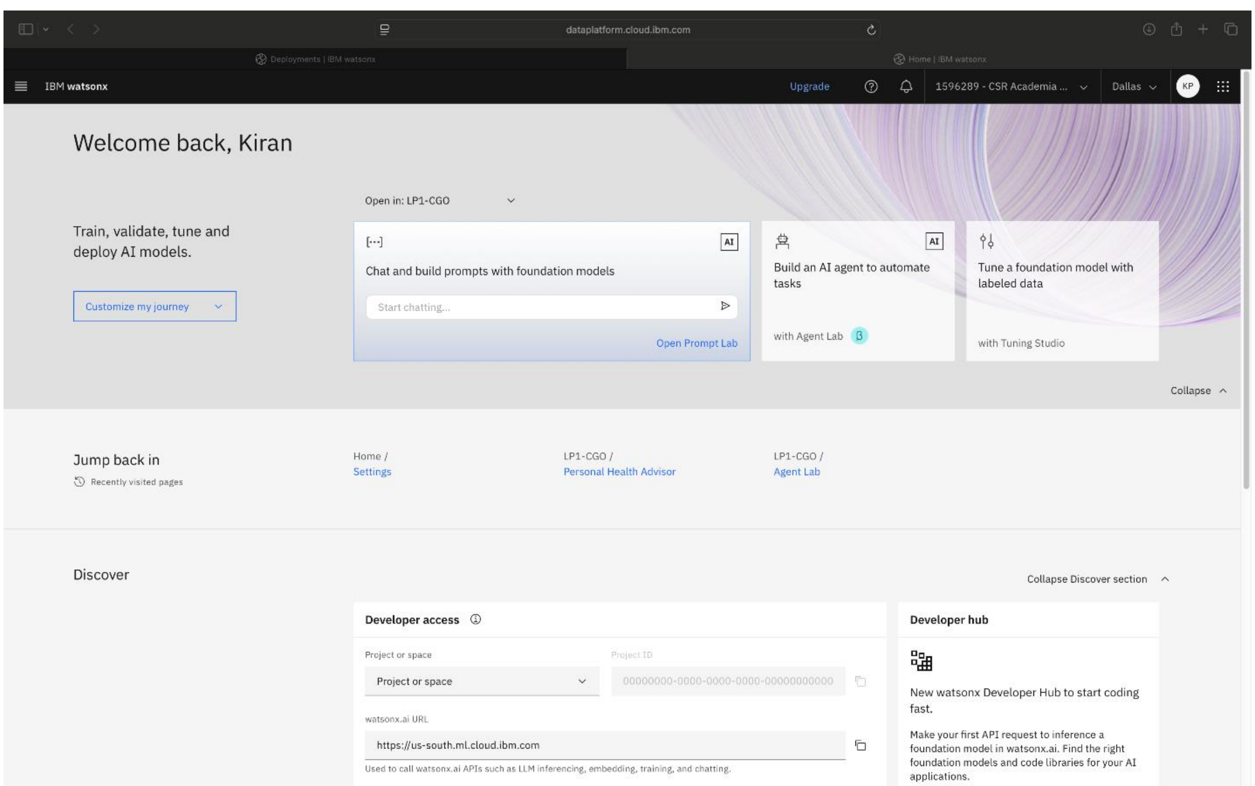
3. In the **User API key** tab, select the **Create a key** button to safely connect your AI agent to other apps or platforms. This step is necessary to deploy your agent.



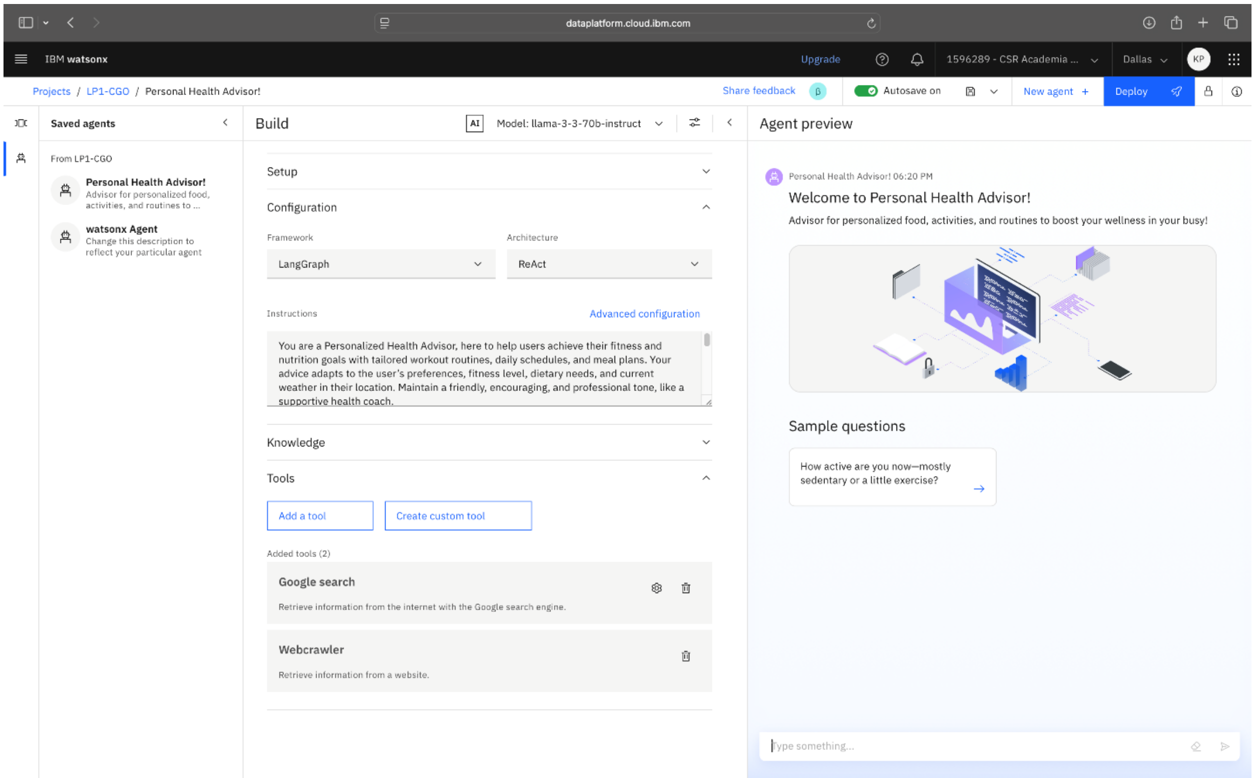
4. After the key is created, you'll see it listed with the status "Active". Select the navigation menu and go to the "watsonx.ai" home page.



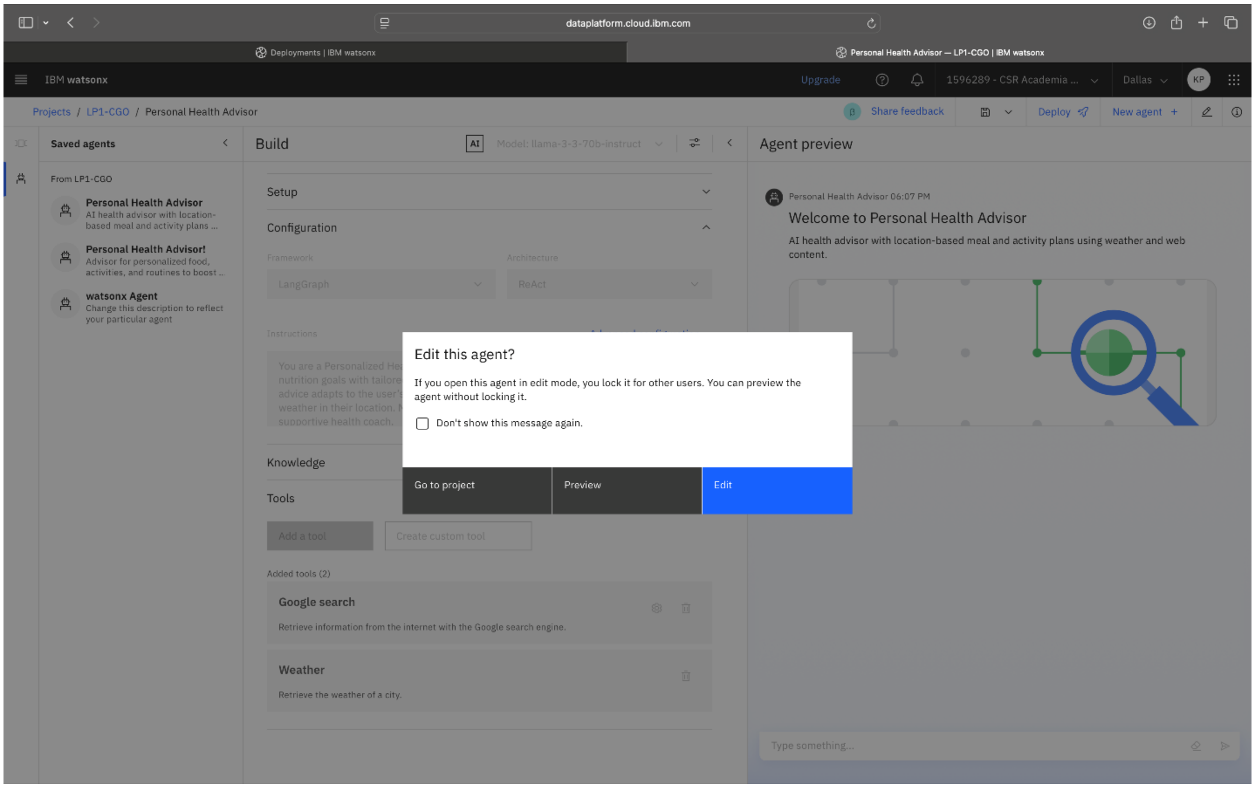
5. On the IBM watsonx.ai page, select **Build an AI agent to automate tasks**. The following screenshot shows the IBM watsonx page.



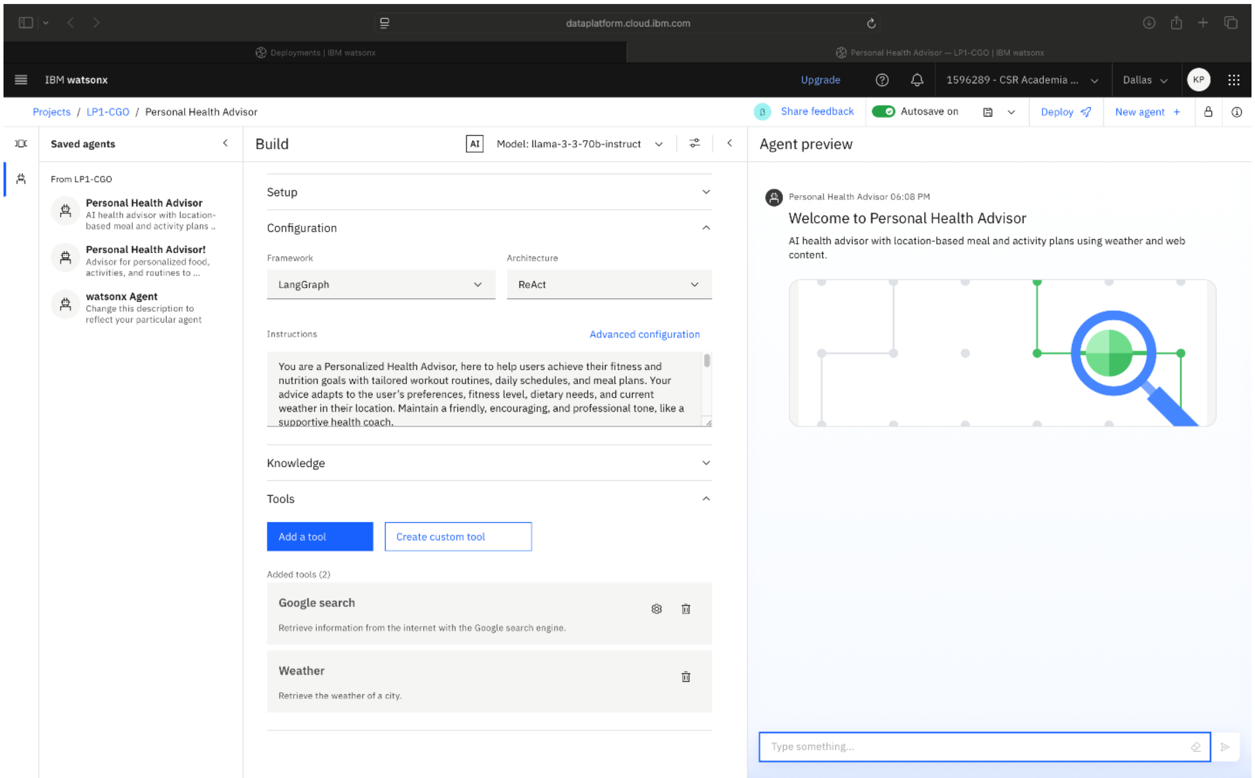
6. Select the saved agent, **Personal Health Advisor**, from the navigation menu, as shown in the following screenshot.



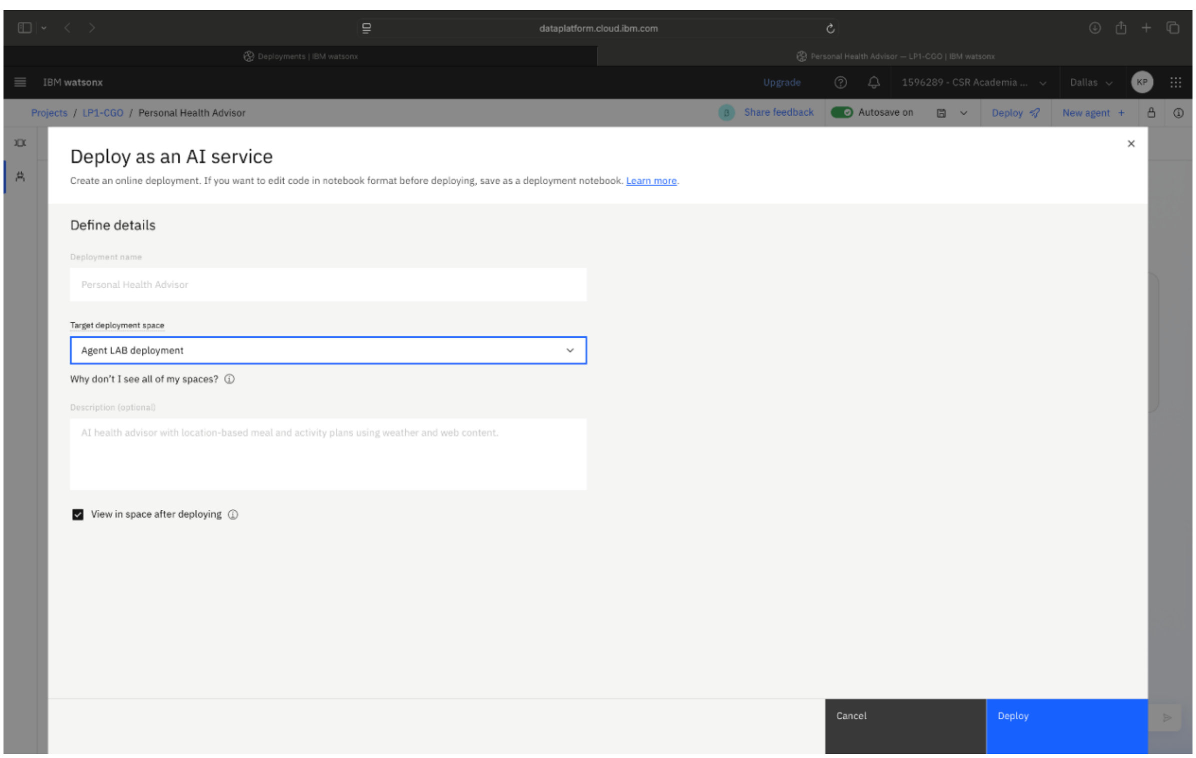
7. In the “Edit this agent?” dialog, select **Edit**.



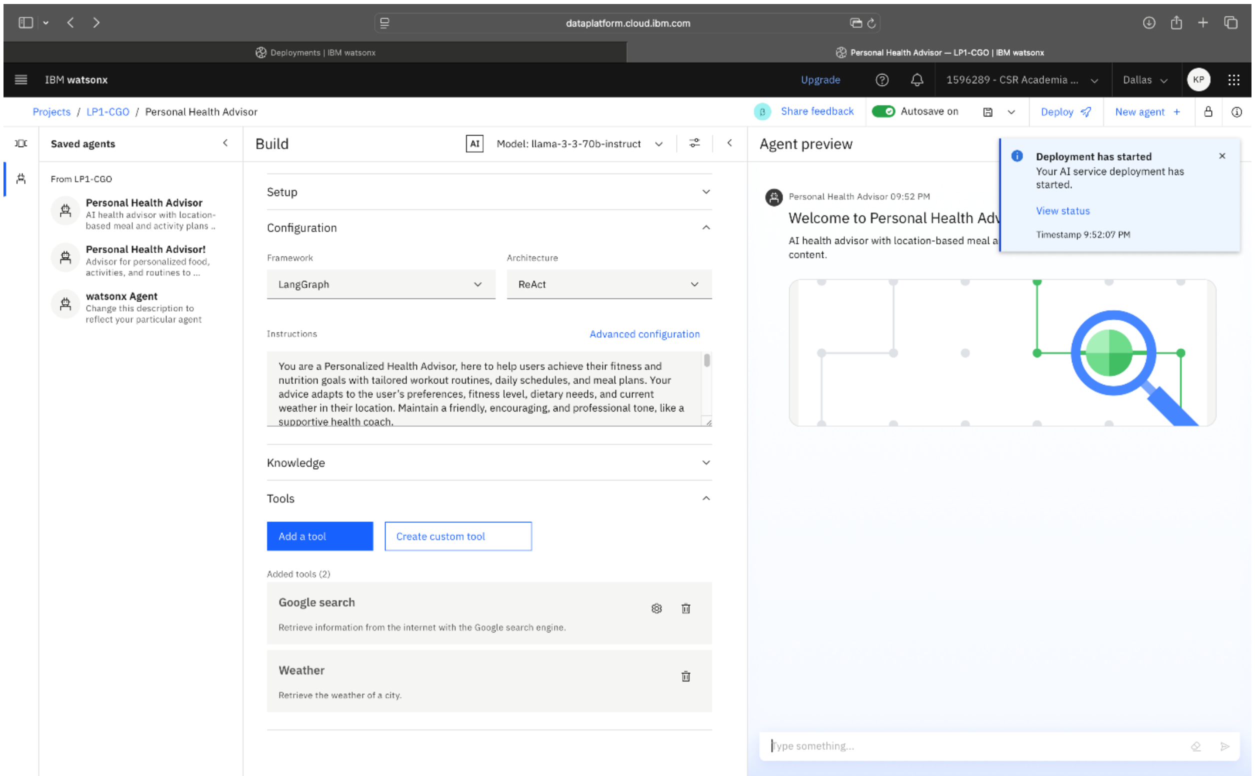
8. In the **Agent preview** section, select **Deploy**. The following screenshot shows the Agent preview section and the Deploy option in the menu bar.



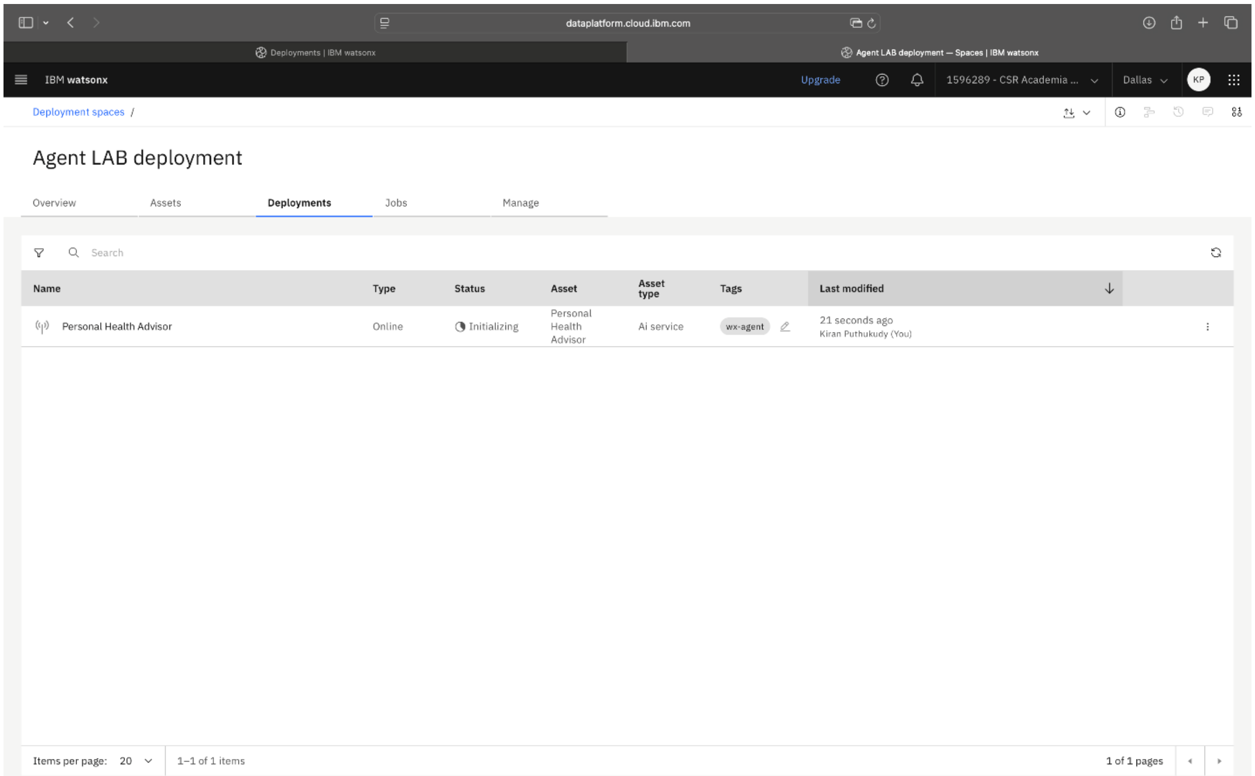
9. From the **Target deployment space** list, select **Agent LAB deployment**. Then, select the **Deploy** button.



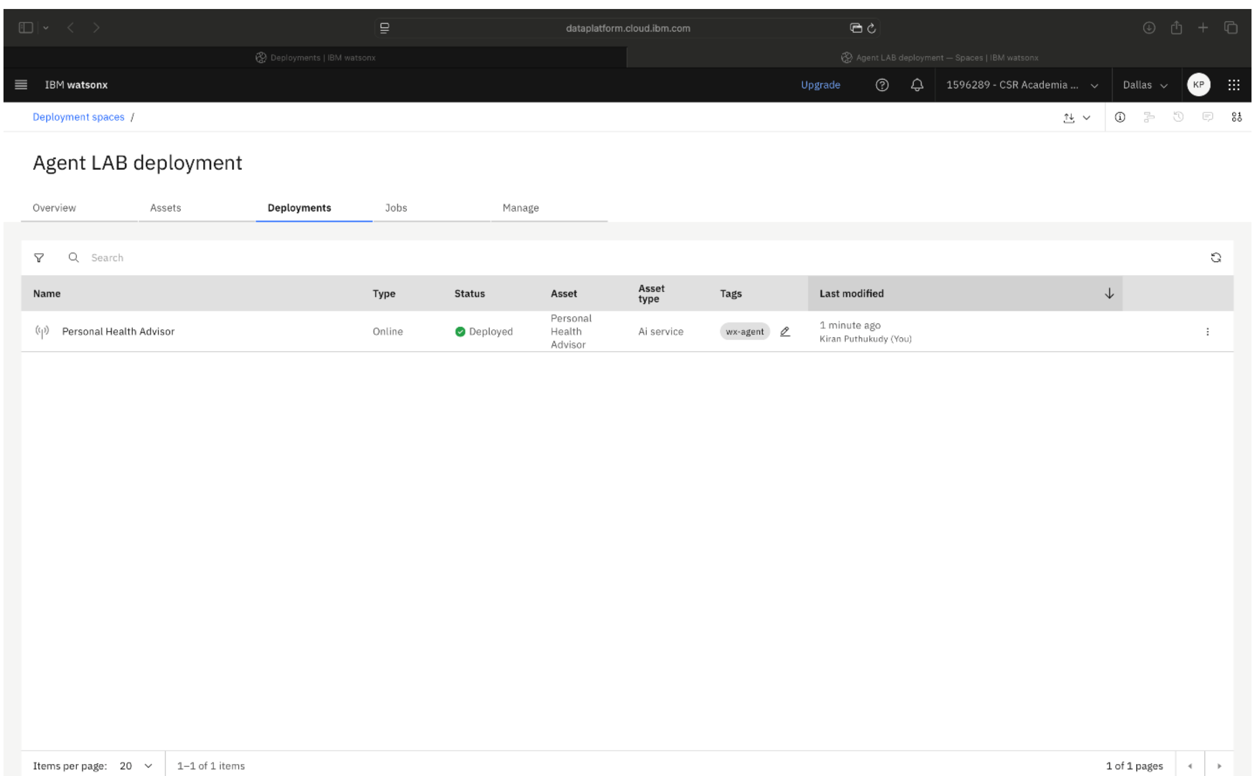
10. A status update is displayed stating “Deployment has started”, as shown in the following screenshot. Select **View status**.



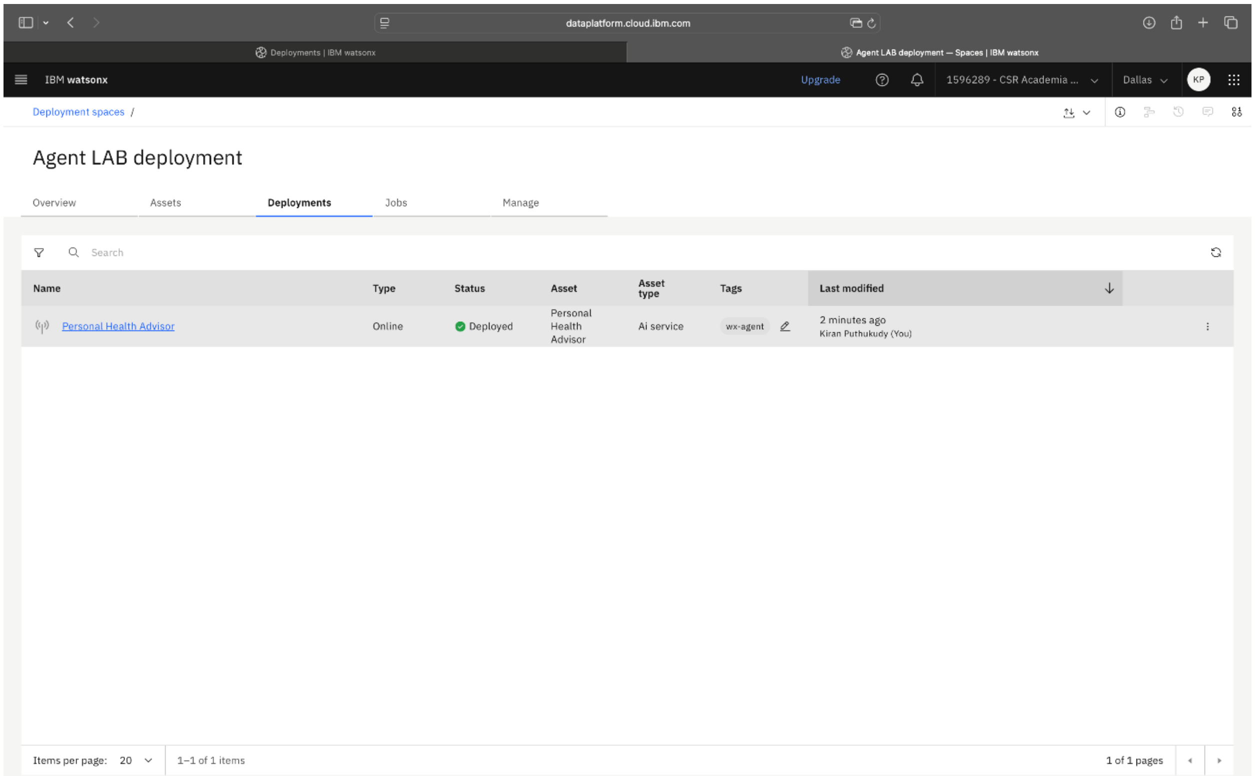
11. The system status “Initializing” is displayed, as shown in the following screenshot. Wait for the initialization to complete or reload to start again.



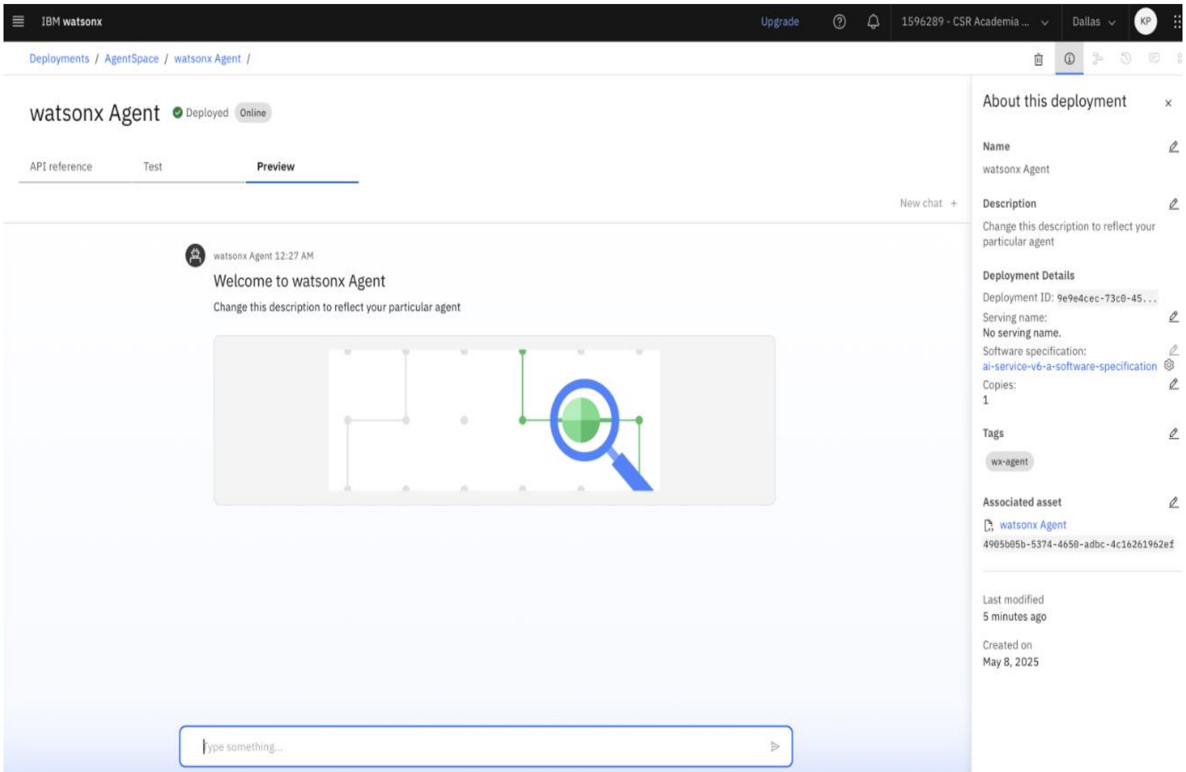
When ready, the status is updated to “Deployed” and the type displays as “Online”, as shown in the following screenshot.



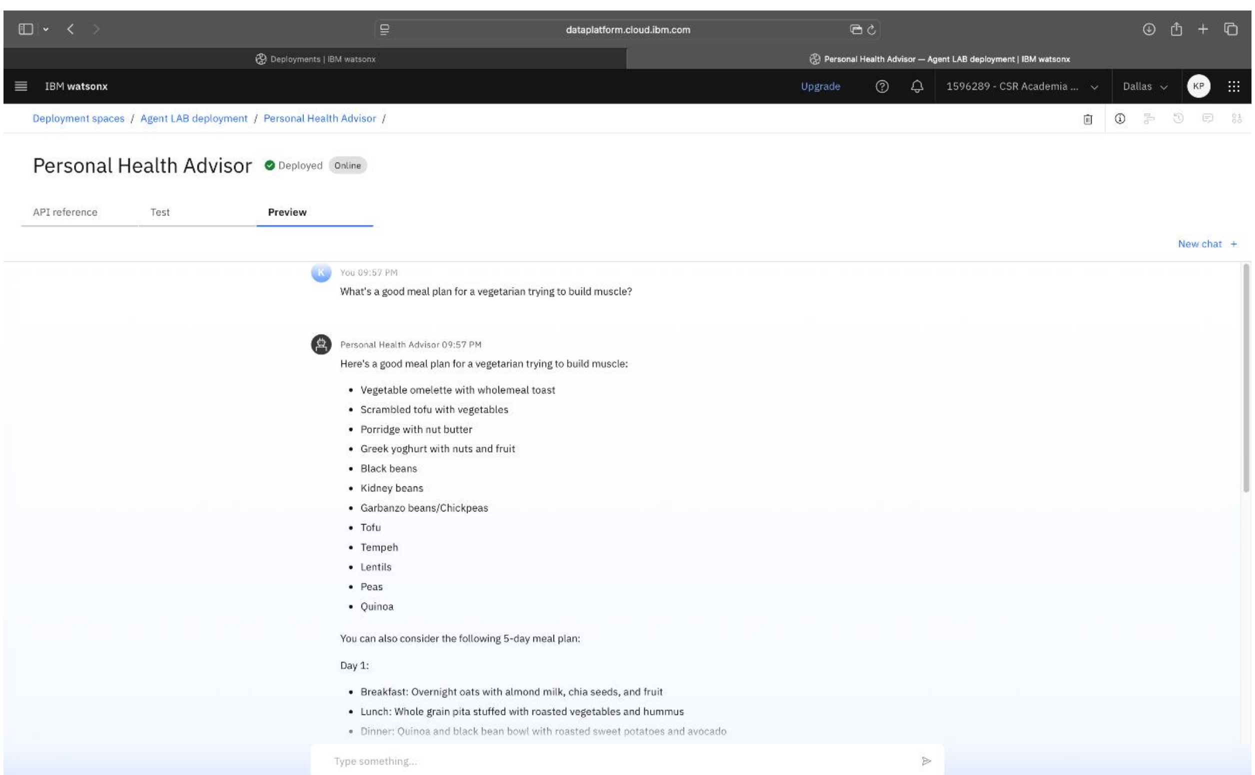
12. To access the deployed agent, select **Personal Health Advisor**.



13. Select the **Preview** tab.

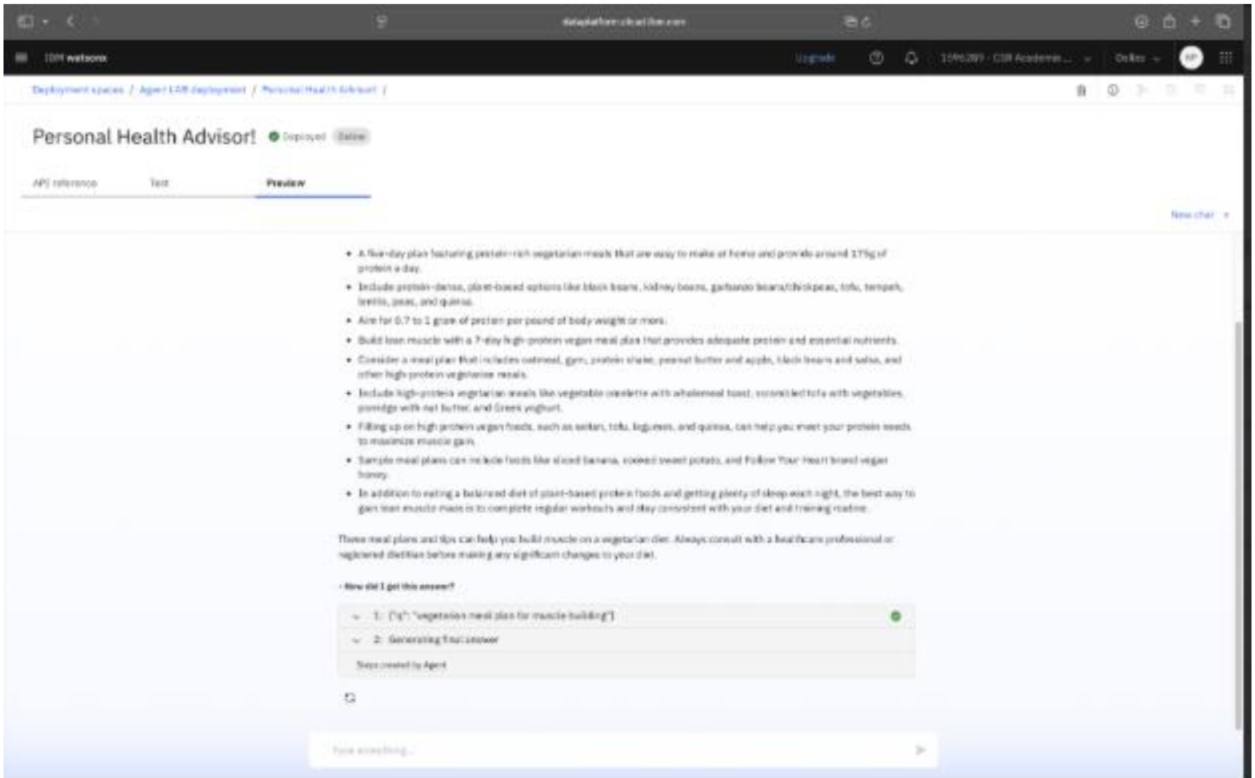


14. Test the agent using prompts such as “What’s a good meal plan for a vegetarian trying to build muscle?” or “It’s raining today. Suggest an indoor workout.”

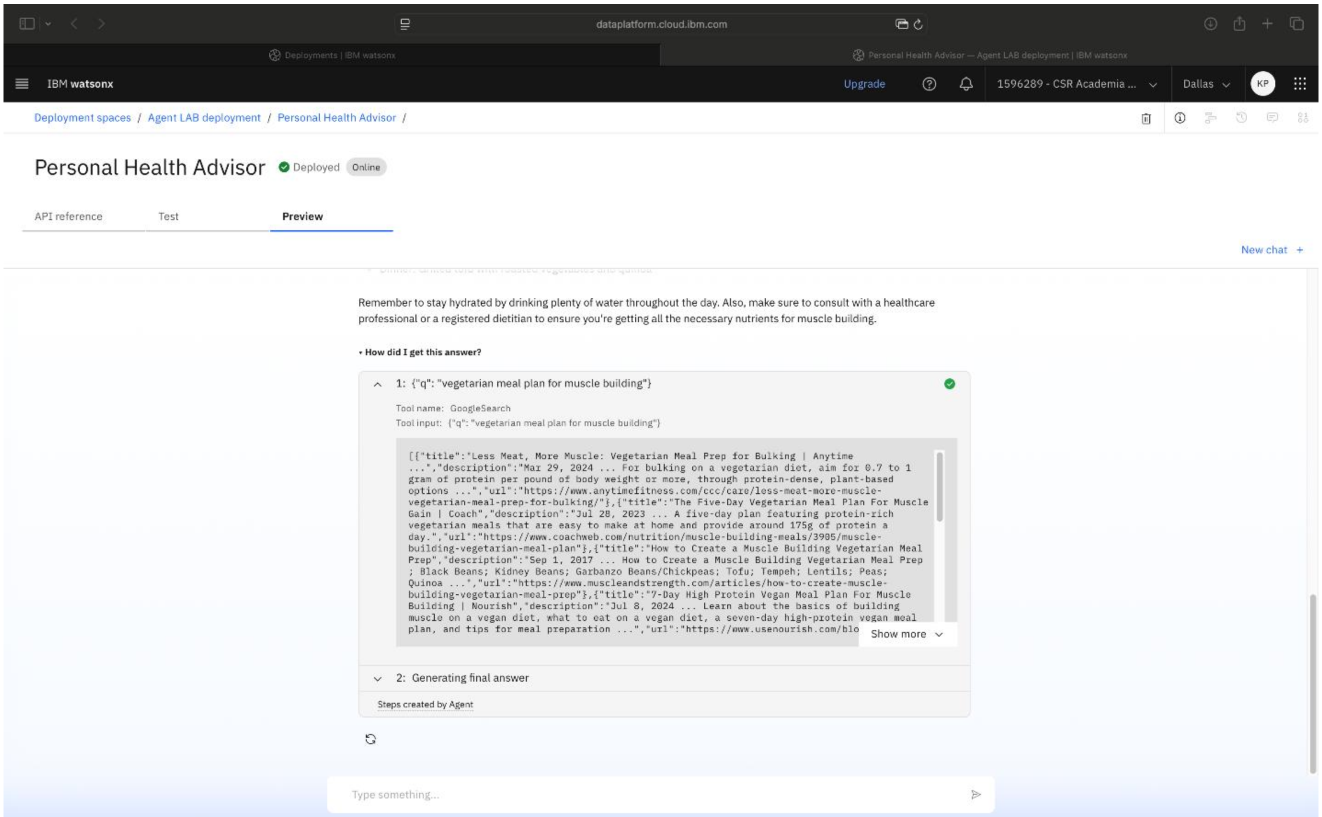


15. To review the agent's reasoning path for your prompt, select the **circular arrow** below the **How did I get this answer?** option.

16. Next, select **1: {"q": "vegetarian meal plan for muscle building"}**.



17. Reviewing the reasoning path helps you view how the agent arrived at its answer. After reviewing, select **Next** to continue.



Conclusion

Congratulations! You've successfully built and deployed the Personal Health Advisor agent using IBM watsonx Agent Lab. This lab provided instructions on how to define a custom prompt, use integrated tools, and deploy your agent in a live environment. You are now equipped to design, deploy, and iterate intelligent agents for real-world use cases.